

Stabilized Space-Time Finite Element Methods for Convection-Diffusion Problems

by

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Abstract

Convection-dominated partial differential equations frequently exhibit severe numerical instabilities when discretized by standard Galerkin methods. This thesis investigates stabilization techniques for the numerical approximation of such equations, with a specific focus on space–time conforming finite element methods.

I analyze three stabilized formulations for the scalar convection–diffusion equation as well as for the vector-valued Oseen problem. For each method, well-posedness and stability are established, and rigorous a priori error estimates are derived. The analysis demonstrates that the stability of the proposed methods remains uniform with respect to the ratio $\|w\|_\infty/\varepsilon$, ensuring that the solution quality does not deteriorate in the convection-dominated regime.

In addition to the theoretical results, a practical implementation strategy is presented. This includes the systematic construction of a four-dimensional space–time finite element solver, addressing the technical challenges of higher-dimensional discretizations and mesh generation.

Chapter 1

Introduction

Solving partial differential equations (PDEs) is one of the central tasks in numerical analysis, as they appear in a wide range of disciplines — from physics and computer science to biology and even economics. PDEs come in many different types, and analytical solutions are rarely available. Therefore, numerous numerical strategies have been developed over the years. The main approaches include the Finite Element Method (FEM), the Finite Difference Method (FDM), and the Finite Volume Method (FVM). Each of these methods has its own advantages and disadvantages, depending on the properties of the underlying PDE. Although the Finite Volume Method (FVM) is widely used in practical Computational Fluid Dynamics applications, the Finite Element Method (FEM) remains a popular and powerful approach for the Navier–Stokes equations due to its flexibility and its suitability for advanced stabilization techniques. Within the FEM framework, various strategies have been developed to overcome difficulties specific to certain PDEs. One major challenge is the convection-dominated regime, which often leads to numerical instabilities and inaccurate solutions. A key idea to address these issues is the Streamline Upwind Petrov–Galerkin (SUPG) method, which was originally introduced by Brooks and Hughes [5] and can be implemented in several different ways. The specific space-time SUPG formulation for the scalar advection diffusion problem analyzed in this work is based on the version proposed in a recent paper published by von Danwitz et al. [7]. While the method was introduced and a numerical convergence study was provided in that paper, the author of this thesis is not aware of any comprehensive mathematical analysis of the method’s error bounds its stability and convergence. This thesis aims to close that gap by presenting a rigorous stability analysis using tools from functional analysis, along with a proof of convergence. Based on the SUPG term for the scalar advection–diffusion problem, I propose an extension in the form of a similar

stabilization term for the vector-valued linearized Navier–Stokes problem. As in the scalar case, I consider a space–time conforming finite element formulation. A convergence proof is provided, along with a numerical study. Due to the space–time conforming nature of the formulation and the three-dimensional spatial domain of the Navier–Stokes equations, the problem is formulated on a four-dimensional space–time domain. However, such four-dimensional space–time discretizations are currently not supported in NGSolve, an open-source finite element software. In particular, it lacks the necessary infrastructure for four-dimensional space–time meshes and finite element spaces. As NGSolve is one of the primary development platforms within the numerical analysis group at the University of Goettingen and constitutes the computational basis of this work, this restriction directly motivates the following developments. I will introduce a method for generating semi-structured four-dimensional space–time meshes, including a systematic construction of the associated simplices.

Mathematical Motivation

Several properties make the space–time finite element framework particularly attractive for the numerical approximation of time-dependent partial differential equations [13, 20]. First, space–time finite elements remain faithful to the fundamental principles of the finite element method and provide a uniform variational framework in which space and time are treated on equal footing. This unified formulation enables a coherent error analysis and naturally allows for adaptive refinement strategies on the combined space–time mesh [8, 20]. Second, the global treatment of space and time can offer advantages with respect to parallelization, as it removes the strictly sequential structure of classical time-stepping schemes and may enable parallel-in-time strategies [22, 13]. Moreover, space–time methods eliminate splitting errors arising from separate spatial and temporal discretizations and permit local temporal refinement without the need for sub-stepping techniques [13, 20]. In particular, simplex-based space–time finite elements yield boundary-conforming space–time meshes and are well suited for problems in which the spatial domain changes topology over time [2].

The Streamline-Upwind Petrov–Galerkin (SUPG) method is chosen in this work due to its robustness and simplicity. SUPG adds a consistent, residual-based stabilization term that acts locally on each element, which enables efficient parallelization and facilitates both implementation and analysis. Furthermore, the stabilization is aligned with the flow direction, avoiding excessive artificial diffusion and preserving sharp solution features. The

SUPG idea is well-known to provide reliable convergence behavior and has been successfully applied to scalar transport equations as well as to linearized Navier–Stokes models.

Limitations/Alternatives

A major drawback of space–time finite element formulations is the increased computational complexity resulting from the additional time dimension [20, 13]. Treating time as an additional coordinate leads to higher-dimensional meshes and a significant growth in the number of degrees of freedom, which results in substantially larger linear systems and increased memory requirements [2, 22]. As the number of unknowns grows rapidly with the dimension, the assembly and solution of the resulting systems may become prohibitively expensive, in particular when direct solvers are employed [20, 1]. This increased numerical effort is one of the main reasons why space–time methods are less common in practical large-scale simulations compared to classical spatial discretizations combined with sequential time-stepping schemes, which often offer a more favorable balance between accuracy and computational efficiency [22, 11]. Furthermore, SUPG represents only one possible stabilization technique, and alternative approaches such as variational multiscale methods, discontinuity-capturing techniques, or interior penalty methods may be more suitable depending on the problem setting [14, 6]. Another limitation of this work is the restriction to Dirichlet boundary conditions; an extension to Neuman or mixed boundary conditions constitutes a natural direction for future research [9, 23].

Organization of the thesis

The whole thesis is divided into two parts. The first part is concerned with the mathematical aspects of the problem. It contains a chapter on mathematical preliminaries, which provides the reader with the necessary background to understand the subsequent theoretical results. Based on this mathematical foundation, a convergence proof is derived for the scalar advection–diffusion problem. Building upon this analysis, the theoretical framework is then extended to a closely related stabilization term for the linearized Navier–Stokes equations.

The second part of the thesis is devoted to the implementation aspects and the construction of the four-dimensional space–time finite element framework, together with the underlying mathematical considerations. Finally, nu-

merical experiments are presented to illustrate and support the theoretical findings

Technical Implementation and Language Support It is noted that generative AI was employed as a supportive tool for the L^AT_EX typesetting, the formatting of complex mathematical expressions, and the refinement of language and grammar throughout this document. The author has manually reviewed, edited, and verified all generated code, mathematical derivations, and linguistic suggestions to ensure scientific accuracy, stylistic consistency, and academic integrity.

Chapter 2

Finite Element Discretization and Stabilization

2.1 Introduction to the Finite Element Method (FEM)

The Finite Element Method (FEM) is one of the most popular and powerful numerical strategies for solving partial differential equations (PDEs). At its core, FEM is built upon two fundamental ideas:

1. Domain decomposition. The computational domain Ω , on which the PDE is defined, is subdivided into a finite number of non-overlapping elements (e.g., triangles, quadrilaterals, tetrahedra, or hexahedra), called *finite elements* (hence the name of the method). This subdivision offers several advantages: it allows complex geometries to be approximated accurately, it enables the efficient resolution of local phenomena, and it provides a convenient and structured representation of the approximate solution. The collection of all elements forms a so-called *admissible triangulation* of Ω .

2. Variational formulation. The second key concept of FEM is to reformulate the PDE as a *variational* or *weak problem*. Instead of enforcing the differential equation pointwise, we multiply it by a *test function* v from a suitable space V and integrate over the entire domain:

$$\int_{\Omega} \mathcal{L}(u) v \, dx = \int_{\Omega} f v \, dx,$$

where $\mathcal{L}$ denotes the differential operator of the PDE and f the right-hand side. By applying integration by parts (if necessary) and incorporating boundary conditions, we obtain the weak formulation:

$$a(u, v) = f(v) \quad \forall v \in V,$$

where $a(\cdot, \cdot)$ is a bilinear form representing the PDE operator and $f(\cdot)$ is a linear functional representing the data.

To approximate this problem numerically, the infinite-dimensional function spaces V are replaced by finite-dimensional subspaces $V_h \subset V$. Here, the previously introduced mesh plays a crucial role: on each element, the discrete space consists of simple basis functions, typically polynomials of degree k . Different variants of the FEM arise from different choices of local approximation spaces and their global assembly. For example, the *continuous Galerkin method* uses globally continuous, piecewise polynomial functions, while *discontinuous Galerkin methods* allow jumps across element boundaries.

The discrete weak problem then reads:

$$\text{Find } u_h \in V_h \text{ such that } a_h(u_h, v_h) = f(v_h) \quad \forall v_h \in Q_h.$$

Here, $a_h(\cdot, \cdot)$ may coincide with the continuous bilinear form $a(\cdot, \cdot)$, but in many cases it incorporates discretization or stabilization terms. In the simplest case, $Q_h = V_h$, but they do not need to be identical. This flexibility leads to various mixed and stabilized formulations.

Once this discrete formulation is established, the finite-dimensional space V_h can be represented by a basis $\{\varphi_i\}_{i=1}^N$. The unknown function u_h is then expressed as a linear combination

$$u_h = \sum_{i=1}^N U_i \varphi_i,$$

where U_i are the coefficients (degrees of freedom). Substituting this expansion into the discrete weak problem yields an algebraic system of equations (linear in the case of linear PDEs)

$$AU = F,$$

where the matrix entries are given by $A_{ij} = a(\varphi_j, \varphi_i)$ and the load vector by $F_i = f(\varphi_i)$. Solving this system provides the coefficients U_i , and hence the approximate finite element solution u_h .

This algebraic formulation—often referred to as the *Galerkin projection*—is the central computational step in the finite element method and forms the foundation for its theoretical analysis and practical implementation.

2.2 Motivation for SUPG

We consider the time-dependent convection-diffusion equation for a scalar field u defined in a domain $\Omega \subset \mathbb{R}^d$:

$$\partial_t u + \mathbf{w} \cdot \nabla u - \varepsilon \Delta u = f \quad \text{in } \Omega \times (0, T], \quad (2.1)$$

where $\mathbf{w}$ is a given velocity field, $\varepsilon > 0$ is the diffusion coefficient, and f is a source term. This equation models the transport and diffusion of a scalar quantity—such as temperature or concentration—within a moving medium, for example, water. The scalar spreads due to two mechanisms: convection, which is governed by the bulk motion of the medium (represented by $\mathbf{w}$), and diffusion, which acts isotropically to smooth out gradients. A particularly challenging class of problems arises in the convection-dominated regime, where $\varepsilon \ll \|\mathbf{w}\|$. In such cases, the solution may develop thin layers in which the gradient of u becomes very large. Depending on the type of PDE and the origin of the layer, the layer thickness scales differently: boundary (exponential) layers typically have a thickness of order $\mathcal{O}(\varepsilon)$, while characteristic (interior) layers—such as those arising from internal sources or discontinuous initial data—exhibit a thickness of order $\mathcal{O}(\sqrt{\varepsilon})$. For the problematic case $\varepsilon \ll h$ standard Galerkin methods struggle to resolve the rapid variation within a single element. As a result, the approximation introduces non-physical oscillations—a manifestation of the Gibbs phenomenon—since smooth finite element basis functions cannot accurately represent steep gradients over coarse meshes. These oscillations often spread globally due to the lack of directional sensitivity in the test space. The SUPG method addresses this by adding a streamline-aligned stabilization term, effectively damping the spurious oscillations without over-smoothing the solution. This makes SUPG particularly effective for problems with unresolved boundary or characteristic layers [15].

2.3 Analysis of the Standard Galerkin Method

To motivate the need for stabilization, let us first consider the standard Galerkin discretization of the homogeneous, time-dependent convection–diffusion problem (2.1). We define the space-time domain $Q := \Omega \times (0, T)$ and let $\mathcal{K}_h$ be an admissible triangulation of Q .

Let $V = H_0^1(Q)$ and let $V_h \subset V$ be a conforming finite element space. The standard Galerkin problem reads: Find $u_h \in V_h$ such that

$$a(u_h, v_h) = F(v_h) \quad \forall v_h \in V_h, \quad (2.2)$$

where the bilinear form $a(\cdot, \cdot)$ is defined as

$$a(u, v) := \varepsilon(\nabla_x u, \nabla_x v)_{L^2(Q)} + (\tilde{w} \cdot \nabla u, v)_{L^2(Q)}, \quad (2.3)$$

with the space-time velocity field $\tilde{w} := (w, 1)$.

By applying the classical Céa strategy, we can derive a quasi-optimality estimate in the energy norm $\|v\|_V := \sqrt{\varepsilon} \|\nabla_x v\|_{L^2(Q)}$. This result is summarized in the following lemma.

Lemma 1 (Céa-type quasi-optimality) *Assume $\varepsilon > 0$ and $\nabla \cdot w = 0$. Let C_P denote the Poincaré constant (i.e., $\|v\|_{L^2(\Omega)} \leq C_P \|\nabla v\|_{L^2(\Omega)}$ for all $v \in V$). The Galerkin solution $u_h \in V_h$ satisfies:*

$$\|u - u_h\|_V \leq \left(1 + \frac{C_P \|\tilde{w}\|_{L^\infty(Q)}}{\varepsilon}\right) \inf_{v_h \in V_h} \|u - v_h\|_V. \quad (2.4)$$

Proof. Step 1: Coercivity. For any $v \in V$, the solenoidal property of the field w implies $\nabla \cdot \tilde{w} = \nabla_x \cdot w + \partial_t(1) = 0$. Using integration by parts, we find $(\tilde{w} \cdot \nabla v, v)_{L^2(Q)} = 0$. Thus, we obtain coercivity:

$$a(v, v) = \varepsilon \|\nabla_x v\|_{L^2(Q)}^2 = \|v\|_V^2 \quad \forall v \in V. \quad (2.5)$$

Step 2: Continuity. Using Cauchy–Schwarz and the Poincaré inequality, we estimate:

$$\begin{aligned} |a(u, v)| &\leq \varepsilon \|\nabla_x u\|_{L^2(Q)} \|\nabla_x v\|_{L^2(Q)} + \|\tilde{w}\|_{L^\infty(Q)} \|\nabla u\|_{L^2(Q)} \|v\|_{L^2(Q)} \\ &\leq \left(1 + \frac{C_P \|\tilde{w}\|_{L^\infty(Q)}}{\varepsilon}\right) \|u\|_V \|v\|_V. \end{aligned}$$

Step 3: Galerkin Orthogonality. By construction, $a(u - u_h, v_h) = 0$ for all $v_h \in V_h$. For any $v_h \in V_h$, we have:

$$\begin{aligned} \|u - u_h\|_V^2 &= a(u - u_h, u - u_h) = a(u - u_h, u - v_h) \\ &\leq \left(1 + \frac{C_P \|\tilde{w}\|_{L^\infty(Q)}}{\varepsilon}\right) \|u - u_h\|_V \|u - v_h\|_V. \end{aligned}$$

Dividing by $\|u - u_h\|_V$ and taking the infimum over V_h yields the result. $\square$

Although this estimate suggests an optimal convergence rate of $\mathcal{O}(h^{k+1})$ (if the true solution has enough regularity) in the L^2 -norm for a fixed ε (see, e.g., [18]), the result is highly problematic in practice. In the convection-dominated regime, where $\varepsilon \ll \|\tilde{w}\|_{L^\infty(Q)}$, the stability constant $\|\tilde{w}\|_{L^\infty(Q)}/\varepsilon$ becomes exceedingly large. Consequently, the error bound deteriorates and provides no meaningful control over the approximation. This lack of robust stability manifests as spurious oscillations in the numerical solution, which motivates the development of the SUPG method.

Chapter 3

Space-Time SUPG Method and Analysis

In this chapter, we state the numerical method mentioned earlier, which was originally proposed in [7], and analyze its convergence properties. In particular, we prove that the associated bilinear form is consistent, coercive, and continuous. Based on these properties, we derive an a priori error estimate and show that $\lim_{h \rightarrow 0} \|u - u_h\|_{L^2(\Omega)} = 0$, that is, the discrete solution converges to the exact solution as the mesh size tends to zero.

3.1 Preliminaries

In this section, we collect several auxiliary results that form the technical foundation of the subsequent convergence analysis. These results can be divided into two categories. First, we introduce geometric assumptions on the triangulation, ensuring uniform regularity of the mesh. Second, we recall analytical tools such as trace inequalities, inverse estimates, and interpolation results, which are essential for establishing stability properties and deriving a priori error estimates.

Geometric Assumptions on the Triangulation

Definition 1 (Triangulation) *Let $\Omega \subset \mathbb{R}^d$ be a bounded Lipschitz domain. A finite collection $\mathcal{T}_h$ of nondegenerate d -simplices T is called a (simplicial) triangulation of Ω if:*

1. $\overline{\Omega} = \bigcup_{T \in \mathcal{T}_h} \overline{T}$,
2. *for any $T, T' \in \mathcal{T}_h$, $T \neq T'$, the intersection satisfies $T \cap T' \in \{\emptyset, \text{a common vertex, edge, or } (d-1)\text{-face}\}$,*

3. each $T \in \mathcal{T}_h$ is the interior of its closure.

For $T \in \mathcal{T}_h$, let $h_T := \text{diam}(T)$ be its diameter and ρ_T the inradius (radius of the largest ball contained in T). We set the mesh size $h := \max_{T \in \mathcal{T}_h} h_T$.

Definition 2 (Shape regularity) A triangulation $\mathcal{T}_h$ is called shape-regular if there exists a constant $\sigma \geq 1$ (independent of h) such that

$$\frac{h_T}{\rho_T} \leq \sigma \quad \forall T \in \mathcal{T}_h.$$

Equivalently: all interior angles are uniformly bounded away from 0 and π (in $d=2$), and the aspect ratio of all simplices is uniformly bounded (all d).

Remark 1 (Affine-equivalent formulation) If each $T \in \mathcal{T}_h$ is the affine image $F_T(\hat{T})$ of a fixed reference simplex $\hat{T}$, then shape regularity is equivalent to a uniform bound on the condition numbers of the Jacobians, i.e. $\kappa((\nabla F_T)) \leq C$ for all $T \in \mathcal{T}_h$ (with C independent of h). This extends verbatim to quadrilateral/hexahedral meshes with bilinear/trilinear mappings.

Definition 3 (Admissible triangulation / shape-regular family) A family $\{\mathcal{T}_h\}_{h \downarrow 0}$ is called admissible if:

1. (Conformity) Each $\mathcal{T}_h$ is a triangulation of Ω in the above sense (no hanging nodes; intersections are full common subsimplices).
2. (Shape regularity) There exists σ independent of h such that $\frac{h_T}{\rho_T} \leq \sigma$ for all $T \in \mathcal{T}_h$.
3. (Mesh resolution) $h := \max_{T \in \mathcal{T}_h} h_T \rightarrow 0$ as $h \downarrow 0$.

If, in addition, there exists $C_{\text{qu}} > 0$ with $\max_{T \in \mathcal{T}_h} h_T \leq C_{\text{qu}} \min_{T \in \mathcal{T}_h} h_T$, the family is called quasi-uniform.

Remark 2 (Role of the geometric assumptions) The above conditions ensure that the family of meshes remains uniformly non-degenerate under refinement. In particular, shape regularity prevents arbitrarily small angles and unbounded aspect ratios of elements. This guarantees that constants appearing in trace inequalities, inverse estimates, and interpolation results remain independent of the mesh size h . Without these assumptions, pathological mesh configurations could lead to a deterioration of stability constants and invalidate the subsequent error analysis.

Analytical estimates and important inequalities

Lemma 2 (Cauchy–Schwarz Inequality) *Let $\Omega \subset \mathbb{R}^d$ be measurable and $u, v \in L^2(\Omega)$. Then*

$$|(u, v)_{L^2(\Omega)}| = \left| \int_{\Omega} u(x) \overline{v(x)} \, dx \right| \leq \|u\|_{L^2(\Omega)} \|v\|_{L^2(\Omega)}.$$

Equality holds if and only if u and v are (almost everywhere) linearly dependent.

Lemma 3 (Trace inequality on Lipschitz domains) *Let $\Omega \subset \mathbb{R}^d$ be a bounded Lipschitz domain. Then the trace operator $\text{tr} : H^1(\Omega) \rightarrow L^2(\partial\Omega)$ is bounded, i.e.,*

$$\|v\|_{L^2(\partial\Omega)} \leq C_{\text{tr}} \|v\|_{H^1(\Omega)} \quad \forall v \in H^1(\Omega).$$

Equivalently, there exists $C > 0$ such that

$$\int_{\partial\Omega} |v|^2 \, ds \leq C \left(\int_{\Omega} |v|^2 \, dx + \int_{\Omega} |\nabla v|^2 \, dx \right).$$

Lemma 4 (Local inverse inequality) *Let $\mathcal{T}_h$ be a shape-regular triangulation of $\Omega \subset \mathbb{R}^d$ and let V_h be a finite element space of piecewise polynomials of fixed degree $r \geq 0$. Then for every element $T \in \mathcal{T}_h$ and every $v_h \in V_h$,*

$$\|\nabla v_h\|_{L^2(T)} \leq C_{\text{inv}} h_T^{-1} \|v_h\|_{L^2(T)}.$$

More generally, for integers $0 \leq s \leq m$,

$$\|D^m v_h\|_{L^2(T)} \leq C_{\text{inv}} h_T^{s-m} \|D^s v_h\|_{L^2(T)}.$$

The constant C_{inv} depends only on the shape-regularity of $\mathcal{T}_h$ (and on the polynomial degree r , but not on h_T or v_h).

Lemma 5 (Local interpolation estimate) *Let $\Omega \subset \mathbb{R}^d$ be a bounded Lipschitz domain and $\mathcal{T}_h$ a shape-regular triangulation into elements T with diameter h_T . Let $r \geq 0$ and define on each element T the local L^2 -projection $\Pi_T : L^2(T) \rightarrow \mathbb{P}_r(T)$ by*

$$\int_T (\Pi_T u) q \, dx = \int_T u q \, dx \quad \forall q \in \mathbb{P}_r(T).$$

Then, for all integers $0 \leq m \leq s \leq r + 1$ and all $u \in H^s(T)$,

$$|u - \Pi_T u|_{H^m(T)} \leq C h_T^{s-m} |u|_{H^s(T)}.$$

The constant $C > 0$ depends only on the polynomial degree r , the spatial dimension d , and the shape-regularity of $\mathcal{T}_h$ (but not on h_T or u).

Remark 3 (Notes on Proofs) *All lemmata presented in this section are standard results in the field. Detailed proofs can be found, for example, in [16].*

Remark 4 (Role of the analytical estimates) *The inequalities and interpolation estimates collected above provide the core analytical tools for the convergence analysis. Trace and inverse inequalities allow us to control boundary and higher-order terms in terms of mesh-dependent norms, which is essential for proving continuity and coercivity of the bilinear form. The local interpolation estimate quantifies the approximation properties of the finite element space and forms the basis for deriving the a priori error bounds.*

3.2 Space-Time-Continuous SUPG Method

Our guiding example is the following time-dependent convection–diffusion equation:

$$\partial_t u - \varepsilon \Delta u + w \cdot \nabla u = f \quad \text{in } \Omega \times (0, T), \quad (3.1)$$

where $\varepsilon > 0$ denotes the diffusion coefficient. The equation is supplemented with Dirichlet boundary conditions

$$u = g \quad \text{on } \partial\Omega \times (0, T), \quad (3.2)$$

and the initial condition

$$u(\cdot, 0) = u_0 \quad \text{in } \Omega. \quad (3.3)$$

Furthermore, we assume the transport field w to be solenoidal, i.e.,

$$\nabla \cdot w = 0 \quad \text{in } \Omega, \quad (3.4)$$

which corresponds to the incompressibility of the flow field.

3.2.1 Bilinear form and spaces

We define the space-time domain $Q := \Omega \times (0, T)$. Let $\mathcal{K}_h$ be an admissible triangulation of Q into space-time elements. With respect to this mesh, we let $V \subset H_0^1(Q)$ be a suitable conforming subspace.

The weak formulation of problem (3.1) then reads: Find $u \in V$ such that

$$a(u, v) = f(v) \quad \forall v \in V,$$

where the bilinear form $a(\cdot, \cdot)$ and the linear functional $f(\cdot)$ are defined as

$$a(u, v) := (\partial_t u, v)_Q + \varepsilon(\nabla u, \nabla v)_Q + (w \cdot \nabla u, v)_Q \quad \text{and} \quad f(v) := (f, v)_Q.$$

Here, $(\cdot, \cdot)_Q$ denotes the L^2 -inner product over the space-time domain Q .

Let $\mathcal{K}_h$ be an admissible, shape-regular triangulation of Q . We define the conforming Lagrange finite element space of polynomial degree $k \geq 1$ by

$$V_h := \{v_h \in H_0^1(Q) : v_h|_K \in \mathcal{P}_k(K) \text{ for all } K \in \mathcal{K}_h\}. \quad (3.5)$$

Then $V_h \subset V$ is a conforming finite element space with respect to the triangulation $\mathcal{K}_h$. The standard Galerkin discretization reads: Find $u_h \in V_h$ such that

$$a(u_h, v_h) = f(v_h) \quad \forall v_h \in V_h.$$

In the SUPG method, we keep the same conforming trial and test space V_h , but modify the variational formulation by adding a stabilization term. More precisely, we seek $u_h \in V_h$ such that

$$A(u_h, v_h) := a(u_h, v_h) + a_{\text{SUPG}}(u_h, v_h) = f(v_h) + f_{\text{SUPG}}(v_h) =: F(v_h) \quad \forall v_h \in V_h.$$

where

$$a_{\text{SUPG}}(v) := \sum_{K \in \mathcal{K}_h} \gamma_K \int_K (\partial_t u - \varepsilon \Delta u + w \cdot \nabla u)(w \cdot \nabla v + \partial_t v) dq$$

and

$$f_{\text{SUPG}}(v) := \sum_{K \in \mathcal{K}_h} \gamma_K \int_K f(w \cdot \nabla v + \partial_t v) dq.$$

The stabilization parameter is defined by

$$\gamma_K = \left(\begin{pmatrix} w \\ 1 \end{pmatrix} \cdot G \begin{pmatrix} w \\ 1 \end{pmatrix} + \left(\frac{C_{\text{inv}}^2 \epsilon}{h^2} \right)^2 \right)^{-1/2}, \quad (3.6)$$

Here, G denotes the so-called metric tensor, which is based on the inverse of the Jacobian transformation and ensures that the method remains invariant with respect to node numbering. For further details we refer to the literature. It is defined as

$$G = \left(\frac{\partial(\xi, \tau)}{\partial(x, t)} \right)^T M \left(\frac{\partial(\xi, \tau)}{\partial(x, t)} \right),$$

where M is a reference matrix that depends on the dimension and the shape of the basis element. For simplices one has

$$M_{d=2} = \frac{1}{\sqrt{3}} \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}, \quad M_{d=3} = \frac{1}{\sqrt[3]{4}} \begin{pmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{pmatrix},$$

$$M_{d=4} = \frac{1}{\sqrt[4]{5}} \begin{pmatrix} 2 & 1 & 1 & 1 \\ 1 & 2 & 1 & 1 \\ 1 & 1 & 2 & 1 \\ 1 & 1 & 1 & 2 \end{pmatrix}.$$

We will later discuss the reasoning behind the choice of the stabilization parameter in more detail. For now, it is important to note that it depends on the mesh size, the trace inverse constant, and the diffusion parameter on the one hand, while on the other hand there is a contribution from the metric tensor, which ensures independence from the node numbering. The stabilizing effect is mainly a result of the symmetric term

$$\int_K (w \cdot \nabla u + \partial_t u) (w \cdot \nabla v + \partial_t v) dq.$$

To sum up, the weak formulation of our problem now reads as: find $u \in V := H_0^1(Q)$ such that:

$$\begin{aligned} A(u, v) &= \int_0^T \int_{\Omega} \left(\partial_t u v + \varepsilon \nabla u \cdot \nabla v + (w \cdot \nabla u) v \right) dx dt \\ &\quad + \sum_{K \in \mathcal{K}_h} \gamma_K \int_K \left(\partial_t u - \varepsilon \Delta u + w \cdot \nabla u \right) (w \cdot \nabla v + \partial_t v) dq \\ &= \int_0^T \int_{\Omega} f v dx dt + \sum_{K \in \mathcal{K}_h} \gamma_K \int_K f (w \cdot \nabla v + \partial_t v) dq \\ &=: F(v) \quad \forall v \in V. \end{aligned} \tag{3.7}$$

3.2.2 Boundary and Initial Conditions

As we have seen before, we are looking for a function in the space $H_0^1(Q)$, which at first sight may seem to contradict the boundary and initial conditions. However, in the case of Dirichlet and initial data, we can circumvent this issue by splitting

$$u = u_0 + u_g,$$

where $u_0 \in H_0^1(Q)$ and u_g is a lifting function (extension) that incorporates the Dirichlet data on the lateral boundary $\partial\Omega \times (0, T)$ and the initial data u_0 at $\{t = 0\} \times \Omega$. Both parts constitute the boundary of the space-time domain Q . Thus, instead of solving

$$A(u, v) = F(v),$$

we solve for the homogeneous part u_0 :

$$A(u_0 + u_g, v) = F(v).$$

Since u_g is known, we can move it to the right-hand side and obtain the modified problem

$$A(u_0, v) = F(v) - A(u_g, v) =: \tilde{F}(v),$$

where we now seek a solution u_0 that vanishes on the entire parabolic boundary of Q .

3.3 Stability and Error Analysis

3.3.1 Consistency

Up to this point, the operator ∇ has always referred to the spatial components only. For the sake of readability in what follows, we will use ∇ to denote the full space-time gradient, while the purely spatial gradient will be written as ∇_x .

Lemma 6 *Let u be the solution of (3.1) with $u \in H^1(Q) \cap H^2(\mathcal{K}_h)$, where*

$$H^2(\mathcal{T}_h) := \{v \in L^2(Q) \mid v|_T \in H^2(K), K \in \mathcal{K}_h\}.$$

Then, the following holds:

$$A(u - u_h, v_h) = 0 \quad \forall v_h \in V_h \subset V.$$

Proof. If u solves (3.1), then it follows that

$$-\varepsilon \Delta_x u + w \cdot \nabla_x u + \partial_t u - f = 0 \quad \text{in } Q.$$

Multiplying by v_h and integrating over the space-time domain $Q = \Omega \times (0, T)$ yields

$$\int_Q (-\varepsilon \Delta_x u + (w, 1) \cdot \nabla u - f) v_h dQ = 0.$$

Integrating the first term by parts gives

$$\int_Q \varepsilon \nabla_x u \cdot \nabla_x v_h dQ - \int_{\partial\Omega \times (0, T)} \varepsilon (\nabla_x u \cdot n) v_h dS dt + \int_Q (w, 1) \cdot \nabla u v_h dQ = \int_Q f v_h dQ.$$

Since $v_h = 0$ on the boundary $\partial\Omega \times (0, T)$, the boundary integral vanishes, and the remaining terms satisfy

$$a(u, v_h) = f(v_h).$$

Moreover, it follows directly from (3.1) that the stabilization terms satisfy

$$a_{\text{SUPG}}(u, v_h) = f_{\text{SUPG}}(v_h).$$

Combining these two equalities, we obtain

$$A(u, v_h) = F(v_h) \quad \forall v_h \in V_h,$$

which proves the lemma. $\square$

3.3.2 Coercivity

To show coercivity on V_h , we introduce a modified discrete norm defined by

$$\| \| v_h \| \|^2 := \varepsilon \|\nabla_x v_h\|_{L^2(Q)}^2 + \sum_{K \in \mathcal{K}_h} \gamma_K \|(w, 1) \cdot \nabla v_h\|_{L^2(K)}^2, \quad v_h \in V_h.$$

Remark 5 (Norm property) *The quantity $\| \| \cdot \|$ defines a norm on V_h . Non-negativity and homogeneity are immediate. To verify definiteness, assume $\| \| v_h \| \| = 0$. This implies $\nabla v_h = 0$ in Q and $(w, 1) \cdot \nabla v_h = 0$ on every element $K \in \mathcal{K}_h$.*

The first condition, $\nabla v_h = 0$, implies that v_h is spatially constant. Since functions in V_h satisfy homogeneous Dirichlet boundary conditions on the lateral boundary of Q , this constant must vanish, and hence $v_h = 0$. Therefore, the expression defines a norm on V_h .

Lemma 7 *For $u_h \in V_h$ there holds*

$$A(u_h, u_h) \geq \frac{1}{2} \| \| u_h \| \|^2, \quad (3.8)$$

provided that for polynomial order $k > 1$ the local stabilization parameter satisfies

$$\gamma_K \leq \frac{h_K^2}{c_{inv}^2 \varepsilon}, \quad K \in \mathcal{K}_h, \quad h_K = \text{diam}(K), \quad (3.9)$$

where c_{inv} is the inverse inequality constant such that

$$\|\Delta_x u_h\|_{L^2(K)} \leq \frac{c_{inv}}{h_K} \|\nabla_x u_h\|_{L^2(K)}.$$

Proof. We begin by evaluating the bilinear form for u_h :

$$\begin{aligned} A(u_h, u_h) &= \varepsilon \|\nabla_x u_h\|_{L^2(Q)}^2 + ((w, 1) \cdot \nabla u_h, u_h)_{L^2(Q)} \\ &\quad + \sum_{K \in \mathcal{K}_h} \gamma_K \int_K (-\varepsilon \Delta_x u_h + (w, 1) \cdot \nabla u_h) ((w, 1) \cdot \nabla u_h) dq. \end{aligned}$$

Using integration by parts on the transport term, we observe that

$$((w, 1) \cdot \nabla u_h, u_h)_{L^2(Q)} = -((w, 1) \cdot \nabla u_h, u_h)_{L^2(Q)} = 0,$$

due to the divergence-free property $\nabla \cdot (w, 1) = 0$ and the vanishing boundary conditions $u_h = 0$ on ∂Q . Thus, the convective term vanishes, and the bilinear form reduces to

$$A(u_h, u_h) = \varepsilon \|\nabla_x u_h\|_{L^2(Q)}^2 + \sum_{K \in \mathcal{K}_h} \gamma_K \int_K (-\varepsilon \Delta_x u_h + (w, 1) \cdot \nabla u_h) ((w, 1) \cdot \nabla u_h) dq.$$

Applying the Cauchy–Schwarz inequality to the stabilization integrand on each element K , we obtain

$$\begin{aligned} \int_K (-\varepsilon \Delta_x u_h + (w, 1) \cdot \nabla u_h) ((w, 1) \cdot \nabla u_h) dq \\ \geq \|(w, 1) \cdot \nabla u_h\|_{L^2(K)}^2 - \varepsilon \|\Delta_x u_h\|_{L^2(K)} \|(w, 1) \cdot \nabla u_h\|_{L^2(K)}. \end{aligned}$$

By employing the inverse inequality $\|\Delta_x u_h\|_{L^2(K)} \leq \frac{c_{inv}}{h_K} \|\nabla_x u_h\|_{L^2(K)}$ and Young's inequality, the second term on the right-hand side can be bounded as follows:

$$-\varepsilon \|\Delta_x u_h\|_{L^2(K)} \|(w, 1) \cdot \nabla u_h\|_{L^2(K)} \geq -\frac{\varepsilon^2 c_{inv}^2}{2h_K^2} \|\nabla_x u_h\|_{L^2(K)}^2 - \frac{1}{2} \|(w, 1) \cdot \nabla u_h\|_{L^2(K)}^2.$$

Substituting this back into the element-wise estimate yields

$$\begin{aligned} \int_K (-\varepsilon \Delta_x u_h + (w, 1) \cdot \nabla u_h) ((w, 1) \cdot \nabla u_h) dq \\ \geq \frac{1}{2} \|(w, 1) \cdot \nabla u_h\|_{L^2(K)}^2 - \frac{\varepsilon^2 c_{inv}^2}{2h_K^2} \|\nabla_x u_h\|_{L^2(K)}^2. \end{aligned}$$

Summing over all elements and inserting this into the expression for $A(u_h, u_h)$, we get

$$A(u_h, u_h) \geq \sum_{K \in \mathcal{K}_h} \left(\left(\varepsilon - \frac{\gamma_K \varepsilon^2 c_{inv}^2}{2h_K^2} \right) \|\nabla_x u_h\|_{L^2(K)}^2 + \frac{\gamma_K}{2} \|(w, 1) \cdot \nabla u_h\|_{L^2(K)}^2 \right).$$

Using the stability condition $\gamma_K \leq \frac{h_K^2}{c_{inv}^2 \varepsilon}$ from (3.8), it follows that $\varepsilon - \frac{\gamma_K \varepsilon^2 c_{inv}^2}{2h_K^2} \geq \frac{\varepsilon}{2}$. We thus arrive at

$$A(u_h, u_h) \geq \frac{1}{2} \left(\varepsilon \|\nabla_x u_h\|_{L^2(Q)}^2 + \sum_{K \in \mathcal{K}_h} \gamma_K \|(w, 1) \cdot \nabla u_h\|_{L^2(K)}^2 \right),$$

which is equivalent to

$$A(u_h, u_h) \geq \frac{1}{2} \|u_h\|^2.$$

This concludes the proof. $\square$

3.3.3 Continuity

To show continuity we are going to introduce an additional triple star norm

$$\begin{aligned} |||u_h|||_*^2 := & \sum_{K \in \mathcal{K}_h} \left[\gamma_K^{-1} \|u_h\|_{L^2(K)}^2 + \gamma_K \|(w, 1) \cdot \nabla u_h\|_{L^2(K)}^2 + \gamma_K \varepsilon^2 \|\Delta_x u_h\|_{L^2(K)}^2 \right] \\ & + \varepsilon \|\nabla_x u_h\|_{L^2(Q)}^2. \end{aligned} \quad (3.10)$$

Remark 6 *Note that $||| \cdot |||_*$ is clearly a norm on V_h as it consists of the coercivity norm plus additional non-negative terms, thereby inheriting the definiteness property. The additional terms, particularly the L^2 -control and the L^2 -norm of the Laplacian, are essential for establishing the continuity of the stabilized bilinear form.*

Lemma 8 (Continuity on $V \times V_h$) *Assume that u is sufficiently smooth, specifically $u \in V \cap H^2(Q)$, such that $\Delta_x u \in L^2(Q)$. Then the stabilized bilinear form $A(\cdot, \cdot)$ is continuous on $(V \cap H^2(Q)) \times V_h$ in the following sense: For all $v_h \in V_h$, there holds*

$$|A(u, v_h)| \leq \sqrt{3} \, |||u|||_* \, |||v_h|||. \quad (3.11)$$

Proof. Inserting the definition of $A(\cdot, \cdot)$ and splitting the terms gives

$$\begin{aligned} A(u, v_h) = & \varepsilon (\nabla_x u, \nabla_x v_h)_{L^2(Q)} + ((w, 1) \cdot \nabla u, v_h)_{L^2(Q)} \\ & + \sum_{K \in \mathcal{K}_h} \gamma_K (-\varepsilon \Delta_x u + (w, 1) \cdot \nabla u, (w, 1) \cdot \nabla v_h)_{L^2(K)}. \end{aligned}$$

Applying integration by parts to the convective term, noting that $u, v_h = 0$ on the relevant parts of the boundary, we have $((w, 1) \cdot \nabla u, v_h)_{L^2(Q)} = -(u, (w, 1) \cdot \nabla v_h)_{L^2(Q)}$. Thus,

$$\begin{aligned} A(u, v_h) = & \varepsilon (\nabla_x u, \nabla_x v_h)_{L^2(Q)} - (u, (w, 1) \cdot \nabla v_h)_{L^2(Q)} \\ & + \sum_{K \in \mathcal{K}_h} \gamma_K (-\varepsilon \Delta_x u + (w, 1) \cdot \nabla u, (w, 1) \cdot \nabla v_h)_{L^2(K)}. \end{aligned}$$

Using the Cauchy–Schwarz inequality on each term and grouping the results:

$$\begin{aligned} |A(u, v_h)| \leq & \sqrt{\varepsilon} \|\nabla_x u\|_{L^2(Q)} \sqrt{\varepsilon} \|\nabla_x v_h\|_{L^2(Q)} \\ & + \sum_{K \in \mathcal{K}_h} \sqrt{\gamma_K} \|(w, 1) \cdot \nabla v_h\|_{L^2(K)} \left(\gamma_K^{-1/2} \|u\|_{L^2(K)} \right. \\ & \left. + \sqrt{\gamma_K} \|(w, 1) \cdot \nabla u\|_{L^2(K)} + \sqrt{\gamma_K} \varepsilon \|\Delta_x u\|_{L^2(K)} \right). \end{aligned}$$

Applying the Cauchy–Schwarz inequality for sums yields

$$\begin{aligned}
|A(u, v_h)| &\leq \left(\varepsilon \|\nabla_x u\|_{L^2(Q)}^2 + \sum_{K \in \mathcal{K}_h} \left[\gamma_K^{-1} \|u\|_{L^2(K)}^2 \right. \right. \\
&\quad \left. \left. + \gamma_K \|(w, 1) \cdot \nabla u\|_{L^2(K)}^2 + \gamma_K \varepsilon^2 \|\Delta_x u\|_{L^2(K)}^2 \right] \right)^{1/2} \\
&\quad \cdot \left(\varepsilon \|\nabla_x v_h\|_{L^2(Q)}^2 + 3 \sum_{K \in \mathcal{K}_h} \gamma_K \|(w, 1) \cdot \nabla v_h\|_{L^2(K)}^2 \right)^{1/2}.
\end{aligned}$$

Recalling the definitions of $\|\cdot\|$ and $\|\cdot\|_*$, we conclude

$$|A(u, v_h)| \leq \sqrt{3} \|\| u \| \|_* \|\| v_h \| \|.$$

□

3.3.4 Céa-type result

Lemma 9 *Let u be the solution of (3.1) with $u \in H^1(Q) \cap H^2(\mathcal{K}_h)$, where the broken Sobolev space is defined as*

$$H^2(\mathcal{K}_h) := \{v \in L^2(Q) \mid v|_K \in H^2(K), K \in \mathcal{K}_h\}.$$

Furthermore, let $u_h \in V_h$ denote the discrete solution satisfying $A(u_h, v_h) = F(v_h)$ for all $v_h \in V_h$. Then, the following Céa-type estimate holds:

$$\|\| u - u_h \| \| \leq (2\sqrt{3} + 1) \inf_{v_h \in V_h} \|\| u - v_h \| \|_* . \quad (3.12)$$

Proof. Applying the properties of coercivity, consistency (Galerkin orthogonality), and continuity, we observe for any $v_h \in V_h$:

$$\begin{aligned}
\frac{1}{2} \|\| u_h - v_h \| \|^2 &\stackrel{(\text{coercivity})}{\leq} A(u_h - v_h, u_h - v_h) \\
&\stackrel{(\text{consistency})}{=} A(u - v_h, u_h - v_h) \\
&\stackrel{(\text{continuity})}{\leq} \sqrt{3} \|\| u - v_h \| \|_* \|\| u_h - v_h \| \|. \quad (3.13)
\end{aligned}$$

Dividing by $\|\| u_h - v_h \| \|^2$ leads to:

$$\|\| u_h - v_h \| \| \leq 2\sqrt{3} \|\| u - v_h \| \|_* .$$

Finally, employing the triangle inequality and noting that $\|\cdot\| \leq \|\cdot\|_*$, we obtain:

$$\begin{aligned} \|u - u_h\| &\leq \|u - v_h\| + \|v_h - u_h\| \\ &\leq \|u - v_h\|_* + 2\sqrt{3} \|u - v_h\|_* \\ &= (2\sqrt{3} + 1) \|u - v_h\|_*. \end{aligned}$$

Taking the infimum over all $v_h \in V_h$ completes the proof. $\square$

3.3.5 A priori error estimate

Lemma 10 (Interpolation error) *Let $Q \subset \mathbb{R}^d$ be a Lipschitz domain and let $\mathcal{K}_h$ be an admissible, quasi-uniform, shape-regular triangulation with local mesh sizes h_K . Let V_h be the Lagrange finite element space of polynomial degree $k > 1$.*

Assume that $u \in H^{k+1}(Q)$ and that the problem is in the convection-dominated regime, i.e.,

$$\|(w, 1)\|_\infty \gg \varepsilon.$$

Let $I_h : C^0(\overline{Q}) \rightarrow V_h$ denote the Lagrange interpolant. Then the following estimate holds:

$$\|u - I_h u\|^2 \leq \|u - I_h u\|_*^2 \leq C h^{2k+1} \|(w, 1)\|_\infty \|u\|_{H^{k+1}(Q)}^2,$$

where the constant $C > 0$ is independent of h and u , but may depend on the shape-regularity of $\mathcal{K}_h$ and the polynomial degree k .

Remark 7 (Lagrange interpolant in four dimensions) *In space-time formulations with $d = 4$, the classical Lagrange interpolant requires point evaluations of u . By Sobolev embedding, $H^s(Q) \hookrightarrow C^0(\overline{Q})$ only holds for $s > d/2 = 2$.*

For polynomial degrees $k \geq 2$, the assumption $u \in H^{k+1}(Q)$ already guarantees $k + 1 > 2$, and hence the interpolant is well-defined.

However, for linear elements ($k = 1$), $u \in H^2(Q)$ is not sufficient in four dimensions to ensure continuity. In this case, a slightly stronger regularity assumption, for instance

$$u \in H^{2+\delta}(Q), \quad \delta > 0,$$

is required to guarantee that the Lagrange interpolant is well-defined.

Proof. By applying inverse estimates to the gradient and Laplacian terms within the dual norm, we obtain:

$$\|u - I_h u\|_*^2 \lesssim \sum_{K \in \mathcal{K}_h} \left(\gamma_K^{-1} + \frac{\gamma_K \|(w, 1)\|_\infty^2 C_{inv}^2}{h_K^2} + \frac{\varepsilon^2 \gamma_K C_{inv}^4}{h_K^4} + \frac{\varepsilon C_{inv}^2}{h_K^2} \right) \|u - I_h u\|_{L^2(K)}^2.$$

Utilizing the standard interpolation result, $\|u - I_h u\|_{L^2(K)}^2 \lesssim h_K^{2k+2} \|u\|_{H^{k+1}(K)}^2$, it follows that:

$$\|u - I_h u\|_*^2 \lesssim \sum_{K \in \mathcal{K}_h} \left(\gamma_K^{-1} + \frac{\gamma_K \|(w, 1)\|_\infty^2 C_{inv}^2}{h_K^2} + \frac{\varepsilon^2 \gamma_K C_{inv}^4}{h_K^4} + \frac{\varepsilon C_{inv}^2}{h_K^2} \right) h_K^{2k+2} \|u\|_{H^{k+1}(K)}^2.$$

In the convection-dominated regime, we choose the stabilization parameter as $\gamma_K \sim h_K / \|(w, 1)\|_\infty$. Given that $h \|(w, 1)\|_\infty \gg h\varepsilon$, we derive:

$$\begin{aligned} \|u - I_h u\|_*^2 &\leq C \sum_{K \in \mathcal{K}_h} \left(\frac{\|(w, 1)\|_\infty}{h_K} + \frac{\|(w, 1)\|_\infty}{h_K} + \frac{\varepsilon^2}{h_K^3 \|(w, 1)\|_\infty} + \frac{\varepsilon}{h_K^2} \right) h_K^{2k+2} \|u\|_{H^{k+1}(K)}^2 \\ &\lesssim \sum_{K \in \mathcal{K}_h} h_K^{2k+1} \|(w, 1)\|_\infty \|u\|_{H^{k+1}(K)}^2. \end{aligned}$$

Finally, we derive the a-priori error estimate. Since the mesh $\mathcal{K}_h$ is quasi-uniform ($h_K \simeq h$), the following global estimate holds:

$$\|u - u_h\|_{L^2(Q)}^2 \leq \|u - u_h\|_*^2 \leq Ch^{2k+1} \|(w, 1)\|_\infty \|u\|_{H^{k+1}(Q)}^2.$$

Taking the square root, we obtain the final convergence rate:

$$\|u - u_h\|_{L^2(Q)} \leq Ch^{k+\frac{1}{2}} \|(w, 1)\|_\infty^{\frac{1}{2}} \|u\|_{H^{k+1}(Q)}.$$

□

It is now clearly visible why the method is advantageous in the convection-dominated case, i.e., when $\varepsilon \rightarrow 0$. The resulting norm estimate no longer diverges as ε tends to zero, in contrast to the standard Galerkin method, where no control is available and the constant C blows up in the convection-dominated regime. This once again highlights the mathematical advantages of the SUPG method compared to the classical Galerkin formulation. This explains the large errors observed for the standard Galerkin method compared to SUPG in such cases. The price we pay for this stability is a loss in our convergence-order which is now only $k+1/2$ instead of $k+1$.

3.3.6 Choice of the stabilization parameter

We briefly justify the choice of the stabilization parameter used in our formulation. On each space-time element K , we define

$$\gamma_K := \left(\begin{pmatrix} w \\ 1 \end{pmatrix}^\top G \begin{pmatrix} w \\ 1 \end{pmatrix} + \left(\frac{C_{\text{inv}}^2 \varepsilon}{h_K^2} \right)^2 \right)^{-1/2},$$

where G denotes the metric tensor associated with the affine mapping from the reference element, h_K the local mesh size, $\varepsilon > 0$ the diffusion parameter, and C_{inv} the inverse inequality constant.

Coercivity constraint. The coercivity proof (7) requires the stabilization parameter to satisfy Assumption (3.9), namely

$$\gamma_K \leq C \frac{h_K^2}{\varepsilon}.$$

This condition originates from controlling the diffusive contribution in the SUPG term. Using the definition of γ_K , we immediately obtain

$$\gamma_K \leq \left(\frac{C_{\text{inv}}^2 \varepsilon}{h_K^2} \right)^{-1} = \frac{h_K^2}{C_{\text{inv}}^2 \varepsilon},$$

which shows that Assumption (3.6) is satisfied uniformly with respect to the mesh size and the diffusion parameter. Hence, the diffusion term in the definition of γ_K is essential for guaranteeing coercivity.

Advection scaling and optimal convergence. The scaling with respect to the advective part is motivated by the a priori error estimate derived in Section (3.8). In the convection-dominated regime, optimal convergence requires that the quantity

$$\gamma_K^{-1} + \gamma_K \frac{\|(w, 1)\|_{L^\infty(K)}^2}{h_K^2}$$

is balanced in such a way that neither term dominates the error bound. This leads to the scaling

$$\gamma_K^{-1} \sim \frac{\|(w, 1)\|_{L^\infty(K)}}{h_K}, \quad \text{or equivalently} \quad \gamma_K \sim \frac{h_K}{\|(w, 1)\|_{L^\infty(K)}}.$$

Such a choice ensures that the stabilization term contributes at the same order as the advective part in the error estimate, thereby yielding the optimal convergence rate in the convection-dominated regime.

Combined scaling. The stabilization parameter can be interpreted in the form

$$\gamma_K = \left(\frac{1}{\tau_{\text{adv}}^2} + \frac{1}{\tau_{\text{diff}}^2} \right)^{-1/2},$$

where τ_{adv} and τ_{diff} denote characteristic advective and diffusive time scales, respectively. The advective contribution ensures that, in the convection-dominated regime, the stabilization is scaled consistently with the a priori error estimate and thus preserves the optimal convergence order. The diffusive contribution, on the other hand, guarantees coercivity and stability in regimes where diffusion is significant.

In this way, the parameter automatically adapts to the local balance between convection and diffusion, providing stability without deteriorating the convergence properties of the method.

Chapter 4

Linearized Navier-Stokes/Oseen equation

$$\left\{ \begin{array}{ll} \partial_t u - \varepsilon \Delta u + (w \cdot \nabla) u - \nabla p = f & \text{in } \Omega \times (0, T), \\ \nabla \cdot u = 0 & \text{in } \Omega \times (0, T), \\ u = u_D & \text{on } \partial\Omega \times (0, T), \\ u(\cdot, 0) = u_0 & \text{in } \Omega. \end{array} \right. \quad (4.1)$$

In chapter 3, we analyzed the time-dependent convection-diffusion equation in the convection-dominated regime. When examining the formulation of the linearized unsteady Navier-Stokes equations, one immediately notices a strong structural similarity between the two problems. The main additional components in the Navier-Stokes system are the incompressibility constraint $\nabla \cdot u = 0$ and the coupling with the pressure gradient term ∇p . As before, we assume the transport field w to be incompressible, i.e., $\nabla \cdot w = 0$. After applying integration by parts, these terms give rise to the characteristic saddle-point structure of the variational formulation.

The weak formulation of (4.1) reads as follows: Given $f \in [L^2(\tilde{Q})]^d$, find the velocity $u \in V_D = \left\{ v \in [H^1(\tilde{Q})]^d : v|_{\partial\Omega \times (0, T)} = u_D, v(\cdot, 0) = u_0 \right\}$ and pressure $p \in Q = L^2(0, T; L_0^2(\Omega))$, such that for all test functions $(v, q) \in V_0 \times Q$:

$$K((u, p), (v, q)) = F(v), \quad (4.2)$$

where the combined bilinear form is defined as

$$K((u, p), (v, q)) := a(u, v) + b(v, p) + b(u, q). \quad (4.3)$$

Here, $a(\cdot, \cdot)$ corresponds to the convection-diffusion operator, and $b(\cdot, \cdot)$ rep-

resents the divergence coupling between velocity and pressure:

$$\begin{aligned} a(u, v) &= \int_0^T \int_{\Omega} (\partial_t u \cdot v + \varepsilon \nabla u : \nabla v + (w \cdot \nabla) u \cdot v) dx dt, \\ b(v, p) &= \int_0^T \int_{\Omega} p(\nabla \cdot v) dx dt. \end{aligned}$$

This structure is typical for mixed formulation problems, whose stability and convergence properties are largely determined by the behavior of the a -block associated with the velocity variable. Since we have already studied this part in detail, many aspects of the forthcoming analysis follow similar lines. However, the same difficulties that arise in the convection-dominated regime for the scalar case persist in the Oseen system. The convective term can lead to numerical instabilities and spurious oscillations if treated with a standard Galerkin formulation. Therefore, we will employ the Streamline Upwind Petrov–Galerkin (SUPG) and Pressure-Stabilizing Petrov–Galerkin (PSPG) techniques to ensure the stability and robustness of the discrete solution. Before proceeding to the specific discretizations of this problem, it is essential to establish several fundamental results for the analysis of mixed formulations. These mathematical tools are primarily required to ensure the existence and uniqueness of a discrete solution. Furthermore, they provide the necessary framework to demonstrate that the proposed stabilized methods are indeed stable and well-posed in the sense of the inf-sup condition.

Remark 8 (Notation for Space-Time Domain) *Throughout this chapter, the symbol $\tilde{Q} := \Omega \times (0, T)$ is used to denote the space-time domain. We distinguish this clearly from the space Q , which represents the pressure space $L^2(0, T; L_0^2(\Omega))$, to avoid any ambiguity in the variational formulation.*

4.1 Theory of mixed Problems

Theorem 1 (Banach–Nečas–Babuška Theorem) *Let X and Y be Hilbert spaces and let*

$$K : X \times Y \rightarrow \mathbb{R}$$

be a bilinear form.

Assume that the following conditions hold:

(i) **Continuity:** *There exists a constant $M > 0$ such that*

$$|K(x, y)| \leq M \|x\|_X \|y\|_Y \quad \text{for all } x \in X, y \in Y.$$

(ii) **Inf-sup condition:** There exists a constant $\beta > 0$ such that

$$\inf_{x \in X \setminus \{0\}} \sup_{y \in Y \setminus \{0\}} \frac{K(x, y)}{\|x\|_X \|y\|_Y} \geq \beta.$$

(iii) **Non-degeneracy in the second argument:**

$$\forall y \in Y : \quad (K(x, y) = 0 \text{ for all } x \in X) \implies y = 0.$$

Then, for any continuous linear functional $F \in Y^*$, there exists a unique solution $x \in X$ such that

$$K(x, y) = F(y) \quad \text{for all } y \in Y.$$

Moreover, the solution satisfies the stability estimate

$$\|x\|_X \leq \frac{1}{\beta} \|F\|_{Y^*}.$$

Remark 9 A proof of this theorem can be found in [9, Theorem 2.36].

Theorem 2 (Brezzi / LBB) Let V and Q be Hilbert spaces, and let $a : V \times V \rightarrow \mathbb{R}$ and $b : V \times Q \rightarrow \mathbb{R}$ be continuous bilinear forms. Given $f \in V'$ and $g \in Q'$, consider the mixed problem: find $(u, p) \in V \times Q$ such that

$$a(u, v) + b(v, p) = f(v) \quad \forall v \in V, \quad (4.4)$$

$$b(u, q) = g(q) \quad \forall q \in Q. \quad (4.5)$$

Assume:

1. **Coercivity on the kernel of b :** There exists $\alpha > 0$ such that

$$a(v, v) \geq \alpha \|v\|_V^2 \quad \forall v \in \ker(b) := \{v \in V : b(v, q) = 0, \forall q \in Q\}.$$

2. **Inf-sup (LBB) condition:** There exists $\beta > 0$ such that

$$\inf_{0 \neq q \in Q} \sup_{0 \neq v \in V} \frac{b(v, q)}{\|v\|_V \|q\|_Q} \geq \beta.$$

Then there exists a unique solution $(u, p) \in V \times Q$ and a constant $C > 0$, depending only on α , β , and the continuity constants of a and b , such that

$$\|u\|_V + \|p\|_Q \leq C (\|f\|_{V'} + \|g\|_{Q'}).$$

The satisfaction of the inf-sup condition at the continuous level is a fundamental requirement for the well-posedness of the problem. However, when moving to a numerical approximation, this property is not automatically inherited by the discrete subspaces. For the standard Galerkin discretization, we must ensure that the pair of finite element spaces (V_h, Q_h) is "compatible" in a way that provides sufficient pressure stability. This leads to the following discrete version of the LBB condition:

Theorem 3 (Discrete Brezzi / LBB for Galerkin methods) *Let V and Q be Hilbert spaces and $a : V \times V \rightarrow \mathbb{R}$, $b : V \times Q \rightarrow \mathbb{R}$ continuous bilinear forms. Let $V_h \subset V$ and $Q_h \subset Q$ be finite-dimensional subspaces. Consider the discrete mixed problem: find $(u_h, p_h) \in V_h \times Q_h$ such that*

$$a(u_h, v_h) + b(v_h, p_h) = f(v_h) \quad \forall v_h \in V_h, \quad (4.6)$$

$$b(u_h, q_h) = g(q_h) \quad \forall q_h \in Q_h. \quad (4.7)$$

Define the discrete kernel ("div-free" subspace)

$$Z_h := \ker(b|_{V_h \times Q_h}) = \{ v_h \in V_h : b(v_h, q_h) = 0 \quad \forall q_h \in Q_h \}.$$

Assume:

1. **Discrete coercivity on Z_h :** *There exists $\alpha_h > 0$ (preferably independent of h) such that*

$$a(v_h, v_h) \geq \alpha_h \|v_h\|_V^2 \quad \forall v_h \in Z_h.$$

2. **Discrete inf-sup (LBB) condition:** *There exists $\beta_h > 0$ (preferably independent of h) such that*

$$\inf_{0 \neq q_h \in Q_h} \sup_{0 \neq v_h \in V_h} \frac{b(v_h, q_h)}{\|v_h\|_V \|q_h\|_Q} \geq \beta_h.$$

Then there exists a unique solution $(u_h, p_h) \in V_h \times Q_h$ and a constant $C > 0$, depending only on α_h , β_h , and the continuity constants of a and b , such that the stability estimate

$$\|u_h\|_V + \|p_h\|_Q \leq C(\|f\|_{V'} + \|g\|_{Q'})$$

holds. Moreover, the discrete solution is quasi-optimal in the product space norm:

$$\|u - u_h\|_V + \|p - p_h\|_Q \leq C \left(\inf_{v_h \in V_h} \|u - v_h\|_V + \inf_{q_h \in Q_h} \|p - q_h\|_Q \right),$$

provided the continuous problem is well-posed and a, b are continuous.

Remark 10 (On the proof) *The above statements are classical results in the theory of saddle-point problems and are commonly referred to as the Babuška–Brezzi theorem. The continuous result was established in Brezzi [3]. The discrete counterpart and the quasi-optimality estimate follow by the same abstract arguments; see, for example, Brezzi and Fortin [4], Chapter II.*

While the discrete inf-sup condition provides the theoretical basis for stability, verifying it directly for a given pair of spaces (V_h, Q_h) can be mathematically challenging. A much more practical and widely used approach to establish this stability is the use of a Fortin operator. This construction allows us to reduce the discrete stability requirement to the existence of a specific bounded projection, as stated in the following criterion:

Theorem 4 (Fortin criterion (discrete inf–sup via Fortin operator))

Let V and Q be Hilbert spaces and let $b : V \times Q \rightarrow \mathbb{R}$ be a continuous bilinear form that satisfies the continuous inf–sup condition

$$\inf_{0 \neq q \in Q} \sup_{0 \neq v \in V} \frac{b(v, q)}{\|v\|_V \|q\|_Q} \geq \beta > 0.$$

Let $V_h \subset V$ and $Q_h \subset Q$ be finite-dimensional subspaces. Then the following are equivalent:

(i) *There is a constant $\tilde{\beta} > 0$, independent of h , such that the discrete inf–sup holds:*

$$\inf_{0 \neq q_h \in Q_h} \sup_{0 \neq v_h \in V_h} \frac{b(v_h, q_h)}{\|v_h\|_V \|q_h\|_Q} \geq \tilde{\beta}.$$

(ii) *There exists a linear operator (Fortin operator) $\Pi_{V_h} : V \rightarrow V_h$ such that*

$$b(v - \Pi_{V_h} v, q_h) = 0 \quad \forall v \in V, \forall q_h \in Q_h, \quad \text{and} \quad \|\Pi_{V_h} v\|_V \leq c \|v\|_V \quad \forall v \in V,$$

with a constant $c > 0$ independent of h .

Moreover, whenever Π_{V_h} exists as above, the discrete inf–sup constant can be chosen as

$$\tilde{\beta} \geq \beta/c,$$

so in particular it is uniform in h if c is uniform in h .

Remark 11 (On the proof) *A proof of the Fortin criterion can be found in Fortin [10].*

Example 1 (LBB-stable velocity–pressure–pairing) *For the following finite element pairings, a Fortin operator can be constructed, and hence the discrete inf–sup condition holds uniformly; see, e.g., [24, 21, 19]. Hence, by the previous theorem, the discrete inf–sup (LBB) condition holds uniformly with a constant independent of the mesh size h :*

- **Taylor–Hood:** $V_h = [\mathbb{P}_{k+1}]^d$, $Q_h = \mathbb{P}_k$ with $k \geq 1$ on simplicial meshes; e.g. $\mathbb{P}_2/\mathbb{P}_1$.
- **MINI element:** $V_h = [\mathbb{P}_1]^d \oplus$ (element bubbles), $Q_h = \mathbb{P}_1$ on triangles/tetrahedra.
- **Scott–Vogelius:** $V_h = [\mathbb{P}_k]^d$, $Q_h = \mathbb{P}_{k-1}$ on suitably refined (e.g. barycentrically refined) meshes; stable for $k \geq 2$ in 2D (and for $k \geq 3$ in 3D under mesh conditions excluding singular vertices).
- **Tensor-product variants (quadrilaterals/hexes):** $V_h = [\mathbb{Q}_k]^d$, $Q_h = \mathbb{Q}_{k-1}$ with $k \geq 2$; in particular $\mathbb{Q}_2/\mathbb{Q}_1$ (Taylor–Hood).

4.2 SUPG Stabilized P1–P1 Formulation

In general, there exist several numerical approaches for the approximation of convection-dominated Oseen problems. In the following, we first introduce a space–time continuous P_1 – P_1 finite element discretization stabilized by a streamline upwind Petrov–Galerkin (SUPG) term. This method can be viewed as a natural continuation of the stabilization strategy developed in Chapter 3.

Remark 12 *Since the P_1 – P_1 velocity–pressure pairing does not satisfy the discrete inf–sup condition (neither for a space only methods see, e.g., [9] nor for space-time elements), a stabilization mechanism is required. This term will be aligned with our SUPG formulation to ensure a stable block system for both velocity and pressure.*

To define the discrete problem, let $\mathcal{K}_h$ be a triangulation of the space-time cylinder Q . We consider a conforming method where the discrete spaces are defined as the intersection of the continuous spaces with the space of piecewise linear functions:

$$\begin{aligned} V_h &:= \{v_h \in V : v_h|_K \in [P_1(K)]^d \quad \forall K \in \mathcal{K}_h\}, \\ Q_h &:= \{q_h \in Q \cap C^0 : q_h|_K \in P_1(K) \quad \forall K \in \mathcal{K}_h\}. \end{aligned}$$

Due to the use of these equal-order P_1/P_1 elements, we define the stabilized discrete bilinear form $K_h(\cdot, \cdot)$. For P_1 elements, second-order derivatives vanish ($\Delta u_h|_K = 0$), allowing us to omit the diffusion term in the stabilization integrals. The discrete problem reads: Find $(u_h, p_h) \in V_h \times Q_h$ such that for all $(v_h, q_h) \in V_h \times Q_h$:

$$K_h((u_h, p_h), (v_h, q_h)) = F_{\text{stab}}(v_h, q_h), \quad (4.8)$$

where the bilinear form is decomposed as:

$$K_h((u_h, p_h), (v_h, q_h)) := a(u_h, v_h) + a_{\text{supg}}(u_h, v_h) + b(v_h, p_h) + b(u_h, q_h) + c(p_h, q_h). \quad (4.9)$$

The individual components are defined using the space-time gradient $(w, 1) \cdot \nabla = \partial_t + w \cdot \nabla_x$ as follows:

$$\begin{aligned} a(u_h, v_h) &:= \varepsilon(\nabla_x u_h, \nabla_x v_h)_{\tilde{Q}} + ((w, 1) \cdot \nabla u_h, v_h)_{\tilde{Q}}, \\ a_{\text{supg}}(u_h, v_h) &:= \sum_{K \in \mathcal{K}_h} \gamma_K((w, 1) \cdot \nabla u_h, (w, 1) \cdot \nabla v_h)_K, \\ b(v_h, p_h) &:= (\nabla_x \cdot v_h, p_h)_Q - \sum_{K \in \mathcal{K}_h} \gamma_K(\nabla_x p_h, (w, 1) \cdot \nabla v_h)_K, \\ b(u_h, q_h) &:= (\nabla_x \cdot u_h, q_h)_Q - \sum_{K \in \mathcal{K}_h} \gamma_K((w, 1) \cdot \nabla u_h, \nabla_x q_h)_K, \\ c(p_h, q_h) &:= \sum_{K \in \mathcal{K}_h} \gamma_K(\nabla_x p_h, \nabla_x q_h)_K. \end{aligned}$$

Note that $b(u_h, q_h)$ is constructed symmetrically to $b(v_h, p_h)$ with respect to the stabilization. The right-hand side is modified to maintain consistency:

$$F_{\text{stab}}(v_h, q_h) = (f, v_h)_Q + \sum_{K \in \mathcal{K}_h} \gamma_K(f, (w, 1) \cdot \nabla v_h + \nabla_x q_h)_K.$$

Remark 13 Our stabilization parameter γ_K is again chosen as defined in (3.6). For the sake of analytical simplicity, we assume a uniform parameter γ_K for all terms. Note that, in general, the parameter could be chosen with a different scaling for the term (c) to potentially further tune the stability properties.

4.2.1 Well-posedness

Lemma 11 (Well-posedness of the stabilized P_1 - P_1 problem)

Assume that $f \in L^2(\tilde{Q})^d$ such that the linear functional $F(v_h) := (f, v_h)_{\tilde{Q}}$ is

bounded on V_h . Let K_h be the stabilized bilinear form and $\mathcal{F}_h$ the stabilized right-hand side defined as

$$\mathcal{F}_h(v_h, q_h) := (f, v_h)_{\tilde{Q}} + \sum_{K \in \mathcal{K}_h} \gamma_K \left((f, (w, 1) \cdot \nabla v_h)_K + (f, \nabla_x q_h)_K \right).$$

Then the discrete problem: Find $(u_h, p_h) \in V_h \times Q_h$ such that

$$K_h((u_h, p_h), (v_h, q_h)) = \mathcal{F}_h(v_h, q_h) \quad \forall (v_h, q_h) \in V_h \times Q_h,$$

admits a unique solution. Moreover, there exists a constant $C > 0$, independent of the mesh size h , such that the following stability estimate holds:

$$\|(u_h, p_h)\|_{W,h} \leq C \|\mathcal{F}_h\|_{(V_h \times Q_h)^*}.$$

Here, $\|\cdot\|_{W,h}$ is the natural product norm on $W_h = V_h \times Q_h$, i.e.,

$$\|(v_h, q_h)\|_{W,h} = (\|v_h\|_V^2 + \|q_h\|_Q^2)^{1/2}.$$

Stability Analysis Strategy The proof of well-posedness for the stabilized Oseen problem will be split into two distinct parts: first, we establish the *inf-sup stability* of the bilinear form K_h by proving its coercivity with respect to a tailored norm; second, we demonstrate the *continuity* of K_h . Together, these properties ensure the existence of a unique discrete solution.

Lemma 12 (Inf-sup stability of K_h) *Let the discrete spaces V_h, Q_h and the stabilized bilinear form K_h be defined as above. Then K_h is stable with respect to the mesh-dependent product norm $\|\cdot\|_{W,h}$, i.e., there exists a constant $\beta > 0$ such that:*

$$\sup_{(v_h, q_h) \in W_h \setminus \{0\}} \frac{K_h((u_h, p_h), (v_h, q_h))}{\|(v_h, q_h)\|_{W,h}} \geq \beta \|(u_h, p_h)\|_{W,h} \quad \forall (u_h, p_h) \in W_h. \quad (4.10)$$

We introduce mesh-dependent norms on the product space $W_h := V_h \times Q_h$ which are tailored to the stabilized formulation. For $v_h \in V_h$ and $q_h \in Q_h$, we set

$$\|v_h\|_{V,h}^2 := \varepsilon \|\nabla_x v_h\|_{L^2(\tilde{Q})}^2 + \sum_{K \in \mathcal{K}_h} \gamma_K \|(w, 1) \cdot \nabla v_h\|_{L^2(K)}^2, \quad (4.11)$$

and

$$\|q_h\|_{Q,h}^2 := \sum_{K \in \mathcal{K}_h} \gamma_K \|\nabla_x q_h\|_{L^2(K)}^2. \quad (4.12)$$

Since $Q_h \subset L_0^2(\Omega)$, constant pressure modes are excluded, and the seminorm induced by the gradient defines a norm on Q_h . Finally, we equip W_h with the product norm

$$\|(v_h, q_h)\|_{W,h}^2 := \|v_h\|_{V,h}^2 + \|q_h\|_{Q,h}^2, \quad (v_h, q_h) \in W_h. \quad (4.13)$$

Proof of Lemma 12. Let $(u_h, p_h) \in W_h$ be arbitrary. We expand the stabilized bilinear form according to its definition:

$$K_h((u_h, p_h), (v_h, q_h)) = a(u_h, v_h) + a_{supg}(u_h, v_h) + b(v_h, p_h) + b(u_h, q_h) + c(p_h, q_h).$$

To show stability, we choose the specific test functions

$$(v_h, q_h) = (u_h, -p_h) \in W_h.$$

With this choice, the divergence coupling and the mixed stabilization terms cancel each other out. For the divergence part, we have:

$$(\nabla_x \cdot u_h, p_h)_{\tilde{Q}} + (\nabla_x \cdot u_h, -p_h)_{\tilde{Q}} = 0,$$

and for the stabilization contributions in $b(\cdot, \cdot)$ and $b(\cdot, \cdot)$, it follows that:

$$\sum_{K \in \mathcal{K}_h} \gamma_K ((\nabla_x p_h, (w, 1) \cdot \nabla u_h)_K - ((w, 1) \cdot \nabla u_h, \nabla_x p_h)_K) = 0.$$

Consequently, the expression simplifies to:

$$K_h((u_h, p_h), (u_h, -p_h)) = a(u_h, u_h) + a_{supg}(u_h, u_h) + \sum_{K \in \mathcal{K}_h} \gamma_K \|\nabla_x p_h\|_{L^2(K)}^2.$$

Recognizing that $a(\cdot, \cdot) + a_{supg}(\cdot, \cdot)$ is identical to the stabilized velocity bilinear form studied in the scalar case, we invoke its coercivity property established in Lemma 7. This yields:

$$a(u_h, u_h) + a_{supg}(u_h, u_h) \geq \frac{1}{2} \left(\varepsilon \|\nabla_x u_h\|_{L^2(Q)}^2 + \sum_{K \in \mathcal{K}_h} \gamma_K \|(w, 1) \cdot \nabla u_h\|_{L^2(K)}^2 \right).$$

Combining these results, we obtain:

$$\begin{aligned} K_h((u_h, p_h), (u_h, -p_h)) &\geq \frac{1}{2} \left(\varepsilon \|\nabla_x u_h\|_{L^2(Q)}^2 + \sum_{K \in \mathcal{K}_h} \gamma_K \|(w, 1) \cdot \nabla u_h\|_{L^2(K)}^2 \right. \\ &\quad \left. + \sum_{K \in \mathcal{K}_h} \gamma_K \|\nabla_x p_h\|_{L^2(K)}^2 \right) \\ &= \frac{1}{2} \|(u_h, p_h)\|_{W,h}^2. \end{aligned}$$

Finally, since $\|(u_h, -p_h)\|_{W,h} = \|(u_h, p_h)\|_{W,h}$, we conclude by the definition of the supremum:

$$\sup_{(v_h, q_h) \in W_h \setminus \{0\}} \frac{K_h((u_h, p_h), (v_h, q_h))}{\|(v_h, q_h)\|_{W,h}} \geq \frac{K_h((u_h, p_h), (u_h, -p_h))}{\|(u_h, -p_h)\|_{W,h}} \geq \frac{1}{2} \|(u_h, p_h)\|_{W,h}.$$

Taking the infimum over $(u_h, p_h) \neq 0$ yields the inf-sup condition with $\beta = 1/2$. $\square$

Lemma 13 (Continuity of K_h on $W \times W_h$) *There exists a constant $M > 0$, independent of the mesh size h , such that the stabilized bilinear form satisfies:*

$$|K_h((u, p), (v_h, q_h))| \leq M \|(u, p)\|_{*,h} \|(v_h, q_h)\|_{*,h} \quad \forall (u, p) \in W, \forall (v_h, q_h) \in W_h.$$

We define the following mesh-dependent norms tailored for the continuity analysis. For the velocity $u \in V$, the norm $\|\cdot\|_{V,*}$ is given by:

$$\|u\|_{V,*}^2 := \sum_{K \in \mathcal{K}_h} \left(\gamma_K^{-1} \|u\|_{L^2(K)}^2 + \gamma_K \|(w, 1) \cdot \nabla u\|_{L^2(K)}^2 \right) + \varepsilon \|\nabla_x u\|_{L^2(Q)}^2. \quad (4.14)$$

The corresponding norm for the pressure $p \in Q$ is defined as:

$$\|p\|_{Q,*}^2 := \sum_{K \in \mathcal{K}_h} \gamma_K \|\nabla_x p\|_{L^2(K)}^2. \quad (4.15)$$

Finally, the product space W is equipped with the combined norm $\|(u, p)\|_{*,h}^2 := \|u\|_{V,*}^2 + \|p\|_{Q,*}^2$.

Proof of Lemma 13. To establish the continuity of K_h , we analyze the individual contributions of the stabilized formulation:

$$K_h((u, p), (v_h, q_h)) = a(u, v_h) + a_{\text{supg}}(u, v_h) + b(v_h, p) + b(u, q_h) + c(p, q_h).$$

Velocity part The continuity of the stabilized velocity terms $a(\cdot, \cdot) + a_{\text{supg}}(\cdot, \cdot)$ has already been established in Lemma 8. Thus, we directly have:

$$|a(u, v_h) + a_{\text{supg}}(u, v_h)| \leq C_a \|u\|_{V,*} \|v_h\|_{V,*}.$$

Coupling term $b(v_h, p)$ Using integration by parts on the divergence term (with $v_h = 0$ on the spatial boundary), we have $b(v_h, p) = (v_h, \nabla_x p)_Q - \sum_{K \in \mathcal{K}_h} \gamma_K (\nabla_x p, (w, 1) \cdot \nabla v_h)_K$. Applying the Cauchy–Schwarz inequality:

$$\begin{aligned} |b(v_h, p)| &\leq \sum_{K \in \mathcal{K}_h} |(v_h, \nabla_x p)_K| + \sum_{K \in \mathcal{K}_h} \gamma_K |(\nabla_x p, (w, 1) \cdot \nabla v_h)_K| \\ &\leq \sum_{K \in \mathcal{K}_h} \left(\gamma_K^{-1/2} \|v_h\|_{L^2(K)} \cdot \gamma_K^{1/2} \|\nabla_x p\|_{L^2(K)} \right) \\ &\quad + \sum_{K \in \mathcal{K}_h} \left(\gamma_K^{1/2} \|\nabla_x p\|_{L^2(K)} \cdot \gamma_K^{1/2} \|(w, 1) \cdot \nabla v_h\|_{L^2(K)} \right). \end{aligned}$$

By the definition of the norms $\|\cdot\|_{V,*}$ and $\|\cdot\|_{Q,*}$, specifically the terms $\gamma_K^{-1} \|v_h\|^2$, $\gamma_K \|(w, 1) \cdot \nabla v_h\|^2$ and $\gamma_K \|\nabla_x p\|^2$, we conclude:

$$|b(v_h, p)| \leq C \|v_h\|_{V,*} \|p\|_{Q,*}.$$

Coupling term $b(u, q_h)$ The second coupling term $b(u, q_h)$ is handled analogously by using the L^2 -duality and the stability of the pressure terms, leading to:

$$|b(u, q_h)| \leq C \|u\|_{V,*} \|q_h\|_{Q,*}.$$

Pressure stabilization $c(p, q_h)$ Finally, the pressure stabilization term is bounded directly by:

$$|c(p, q_h)| = \left| \sum_{K \in \mathcal{K}_h} \gamma_K (\nabla_x p, \nabla_x q_h)_K \right| \leq \|p\|_{Q,*} \|q_h\|_{Q,*}.$$

Combining these estimates via the triangle inequality and the product norm definition completes the proof. $\square$

Proof of Well-posedness. The proof follows from the established properties of the stabilized bilinear form. According to Lemma 8, K_h is continuous on $W_h \times W_h$ with respect to the mesh-dependent norm $\|\cdot\|_{*,h}$. Furthermore, Lemma 12 provides the inf–sup condition (or coercivity-driven stability):

$$\inf_{(u_h, p_h) \in W_h \setminus \{0\}} \sup_{(v_h, q_h) \in W_h \setminus \{0\}} \frac{K_h((u_h, p_h), (v_h, q_h))}{\|(u_h, p_h)\|_{*,h} \|(v_h, q_h)\|_{*,h}} \geq \beta,$$

where the stability constant is established as $\beta = 1/2$.

Since W_h is a finite-dimensional space, these conditions ensure that the assumptions of the Banach–Nečas–Babuška theorem (cf. 1) are satisfied.

Consequently, the discrete problem admits a unique solution $(u_h, p_h) \in W_h$. Moreover, the stability estimate holds:

$$\|(u_h, p_h)\|_{*,h} \leq \frac{1}{\beta} \|\mathcal{F}_h\|_{W_h^*} = 2\|\mathcal{F}_h\|_{W_h^*},$$

which completes the proof. $\square$

4.2.2 A Priori Error Analysis

In this section, we derive an a priori error estimate of Céa type for the stabilized P_1 – P_1 formulation. The first step in this analysis is to establish that the stabilized bilinear form K_h is consistent with the continuous problem.

Lemma 14 (Consistency) *Let (u, p) be the exact solution of the continuous Oseen problem 4.2, satisfying the regularity*

$$u \in V \cap [H^2(\tilde{Q})]^d, \quad p \in Q \cap H^1(\tilde{Q}),$$

such that all terms in the stabilized formulation are well-defined. Then, the pair (u, p) satisfies the discrete variational identity

$$K_h((u, p), (v_h, q_h)) = \mathcal{F}_h(v_h, q_h) \quad \forall (v_h, q_h) \in V_h \times Q_h, \quad (4.16)$$

where K_h is the stabilized bilinear form and $\mathcal{F}_h$ denotes the stabilized right-hand side defined in the previous section.

Proof. The strong form of the momentum equation reads

$$-\varepsilon \Delta_x u + (w, 1) \cdot \nabla u - \nabla p = f \quad \text{in } Q, \quad (4.17)$$

and the incompressibility constraint is $\nabla_x \cdot u = 0$ in Q .

Multiplying the momentum equation (4.17) by a test function $v_h \in V_h$, integrating over Q , and applying integration by parts to the diffusion term yields:

$$\varepsilon(\nabla_x u, \nabla_x v_h)_Q + ((w, 1) \cdot \nabla u, v_h)_Q - (\nabla_x p, v_h)_Q = (f, v_h)_Q.$$

Using integration by parts on the pressure term (assuming $v_h = 0$ on the boundary), we have $-(\nabla_x p, v_h)_Q = (p, \nabla_x \cdot v_h)_Q$. Thus, we obtain the Galerkin identity:

$$\varepsilon(\nabla_x u, \nabla_x v_h)_Q + ((w, 1) \cdot \nabla u, v_h)_Q + (\nabla_x \cdot v_h, p)_Q = (f, v_h)_Q. \quad (4.18)$$

Moreover, multiplying the incompressibility constraint by $q_h \in Q_h$ gives $(\nabla_x \cdot u, q_h)_Q = 0$.

Next, we incorporate the stabilization terms. Define the residual

$$R(u, p) := -\varepsilon \Delta_x u + (w, 1) \cdot \nabla u - \nabla p - f.$$

Since $R(u, p) = 0$ in Q for the exact solution, the following stabilization contributions vanish for any $\gamma_K > 0$:

$$\sum_{K \in \mathcal{K}_h} \gamma_K (R(u, p), (w, 1) \cdot \nabla v_h)_K = 0, \quad \sum_{K \in \mathcal{K}_h} \gamma_K (R(u, p), \nabla_x q_h)_K = 0.$$

Combining the Galerkin identity with the vanishing stabilization terms, we define the "fully consistent" bilinear form $\tilde{K}_h$. For the exact solution (u, p) , it holds:

$$\tilde{K}_h((u, p), (v_h, q_h)) = K_h((u, p), (v_h, q_h)) + \sum_{K \in \mathcal{K}_h} \gamma_K (-\varepsilon \Delta_x u, (w, 1) \cdot \nabla v_h + \nabla_x q_h)_K.$$

As we have shown, $\tilde{K}_h((u, p), (v_h, q_h)) = \mathcal{F}_h(v_h, q_h)$.

However, since we use P_1 finite elements for the discrete solution u_h , the term $\Delta_x u_h$ vanishes identically on each element K . Thus, the stabilization terms in K_h for the discrete solution are:

$$\sum_{K \in \mathcal{K}_h} \gamma_K (-\varepsilon \underbrace{\Delta_x u_h}_{=0} + (w, 1) \cdot \nabla u_h - \nabla_x p_h, (w, 1) \cdot \nabla v_h + \nabla_x q_h)_K.$$

This implies that for our specific choice of V_h , the bilinear form K_h naturally satisfies the relation $K_h((u_h, p_h), (v_h, q_h)) = \mathcal{F}_h(v_h, q_h)$.

This establishes that both the continuous solution (u, p) and the discrete solution (u_h, p_h) satisfy the same stabilized variational identity for all test functions in the discrete space. Consequently, by subtracting the discrete problem from the consistency identity, we obtain the **Galerkin orthogonality**:

$$K_h((u - u_h, p - p_h), (v_h, q_h)) = 0 \quad \forall (v_h, q_h) \in V_h \times Q_h. \quad (4.19)$$

This property is fundamental for the subsequent energy-norm error analysis. $\square$

Theorem 5 (Céa-type a priori error estimate) *Let $(u, p) \in V \times Q$ be the exact solution and $(u_h, p_h) \in W_h$ be the discrete solution of the stabilized P_1 - P_1 formulation. Suppose the stabilized bilinear form K_h is continuous with*

constant $M > 0$ (Lemma 13) and satisfies the inf-sup condition with constant $\beta > 0$ (Lemma 12). Under the assumption of consistency (Lemma 6), the discretization error satisfies:

$$\|(u - u_h, p - p_h)\|_{*,h} \leq \left(1 + \frac{M}{\beta}\right) \inf_{(v_h, q_h) \in W_h} \|(u - v_h, p - q_h)\|_{*,h}.$$

Proof. Let $(w_h, r_h) \in W_h$ be an arbitrary discrete pair. We split the error into an approximation error η and a discrete error e_h using the triangle inequality:

$$\|(u - u_h, p - p_h)\|_{*,h} \leq \underbrace{\|(u - w_h, p - r_h)\|_{*,h}}_{=:\|\eta\|_{*,h}} + \underbrace{\|(w_h - u_h, r_h - p_h)\|_{*,h}}_{=:\|e_h\|_{*,h}}.$$

To bound the discrete error $e_h \in W_h$, we utilize the inf-sup stability (Lemma 12):

$$\beta \|e_h\|_{*,h} \leq \sup_{(v_h, q_h) \in W_h \setminus \{0\}} \frac{K_h(e_h, (v_h, q_h))}{\|(v_h, q_h)\|_{*,h}}.$$

Exploiting the linearity of K_h and the Galerkin orthogonality established in the previous section, $K_h((u - u_h, p - p_h), \cdot) = 0$, we can rewrite the term in the numerator as:

$$\begin{aligned} K_h(e_h, (v_h, q_h)) &= K_h((w_h - u + u - u_h, r_h - p + p - p_h), (v_h, q_h)) \\ &= K_h((w_h - u, r_h - p), (v_h, q_h)) + \underbrace{K_h((u - u_h, p - p_h), (v_h, q_h))}_{=0}. \end{aligned}$$

Applying the continuity of K_h (Lemma 13) to the remaining term yields:

$$|K_h(e_h, (v_h, q_h))| \leq M \|(u - w_h, p - r_h)\|_{*,h} \|(v_h, q_h)\|_{*,h}.$$

Substituting this back into the stability estimate and canceling the supremum over the test functions, we obtain:

$$\|e_h\|_{*,h} \leq \frac{M}{\beta} \|(u - w_h, p - r_h)\|_{*,h}.$$

Combining this with the triangle inequality gives:

$$\|(u - u_h, p - p_h)\|_{*,h} \leq \left(1 + \frac{M}{\beta}\right) \|(u - w_h, p - r_h)\|_{*,h}.$$

The assertion follows by taking the infimum over all $(w_h, r_h) \in W_h$. $\square$

Theorem 6 (A priori error estimate) *Assume that the mesh is shape-regular and quasi-uniform, and that the stabilization parameters satisfy $\gamma_K \simeq h_K$. Let (u, p) be the exact solution with $u \in [H^2(\tilde{Q})]^d$ and $p \in H^1(\tilde{Q})$. Then there exists a constant $C > 0$, independent of h , such that the error of the discrete solution $(u_h, p_h) \in W_h$ satisfies:*

$$\|(u - u_h, p - p_h)\|_{*,h}^2 \leq Ch^3 \|(w, 1)\|_\infty \|u\|_{H^2(Q)}^2 + Ch^3 \|p\|_{H^1(Q)}^2.$$

Proof. Let $I_h : V \rightarrow V_h$ and $J_h : Q \rightarrow Q_h$ denote the Lagrange interpolation operators associated with the finite element spaces V_h and Q_h , respectively. By the Céa-type estimate, the error is bounded by the interpolation error of the exact solution. We choose $v_h = I_h u$ and $q_h = J_h p$.

Remark 14 (Regularity requirement in four space–time dimensions)

In the space–time setting $Q \subset \mathbb{R}^4$, the Lagrange interpolants $I_h u$ and $J_h p$ require point evaluations. By Sobolev embedding, $H^s(Q) \hookrightarrow C^0(\bar{Q})$ only holds for $s > 2$. Hence, in order for the nodal interpolation operators to be well-defined, we assume in the 4d case

$$u \in H^{2+\delta}(Q), \quad p \in H^{2+\delta}(Q), \quad \delta > 0.$$

Velocity error The velocity part of the norm $\|u - I_h u\|_{V,*}$ is directly covered by the result from Lemma 9. For P_1 -elements ($k = 1$), we obtain:

$$\|u - I_h u\|_{V,*}^2 \leq Ch^{2(1)+1} \|(w, 1)\|_\infty \|u\|_{H^2(Q)}^2 = Ch^3 \|(w, 1)\|_\infty \|u\|_{H^2(Q)}^2.$$

Taking the square root gives the $O(h^{3/2})$ contribution.

Pressure interpolation error For the pressure interpolant $J_h p$, we use the element-wise estimate $\|\nabla_x(p - J_h p)\|_{L^2(K)} \leq Ch_K |p|_{H^2(K)}$. Given the scaling $\gamma_K \simeq h_K / \|w\|_K$, we obtain:

$$\begin{aligned} \|p - J_h p\|_{Q,*}^2 &= \sum_{K \in \mathcal{K}_h} \gamma_K \|\nabla_x(p - J_h p)\|_{L^2(K)}^2 \\ &\leq \sum_{K \in \mathcal{K}_h} C \frac{h_K}{\|w\|_K} \left(h_K^2 |p|_{H^2(K)}^2 \right) \\ &= C \sum_{K \in \mathcal{K}_h} \frac{h_K^3}{\|w\|_K} |p|_{H^2(K)}^2 \leq C \frac{h^3}{\|w\|_{\min}} |p|_{H^2(Q)}^2. \end{aligned}$$

Combining the velocity and pressure contributions yields the desired squared estimate, where the h^3 -terms dominate in the convection-dominated regime.

□

4.3 Stabilized P_2 – P_1 Space-Time Formulation

We consider a stabilized P_2 – P_1 discretization for the space–time Oseen problem. Let the discrete spaces be defined as:

$$\begin{aligned} V_h &:= \{v_h \in V : v_h|_K \in [P_2(K)]^d \quad \forall K \in \mathcal{K}_h\}, \\ Q_h &:= \{q_h \in Q \cap C^0 : q_h|_K \in P_1(K) \quad \forall K \in \mathcal{K}_h\}. \end{aligned}$$

where the pressure space is restricted to functions with zero mean:

$$Q := \left\{ q \in L^2(\tilde{Q}) : \int_{\tilde{Q}} q \, d(x, t) = 0 \right\}.$$

This restriction is physically motivated, as the pressure in the Oseen equations is determined only up to an additive constant.

4.3.1 The Stabilized Bilinear Form

For the pairs $(u, p), (v, q) \in V_h \times Q_h$, we define the stabilized bilinear form:

$$K_h((u, p), (v, q)) := A(u, v) + A_{\text{supg}}(u, v) + B(v, p) + B(u, q), \quad (4.20)$$

with the individual components given by:

$$\begin{aligned} A(u, v) &:= \varepsilon(\nabla_x u, \nabla_x v)_Q + ((w, 1) \cdot \nabla u, v)_Q, \\ A_{\text{supg}}(u, v) &:= \sum_{K \in \mathcal{T}_h} \gamma_K ((w, 1) \cdot \nabla u - \varepsilon \Delta_x u, \Pi_K((w, 1) \cdot \nabla v))_K, \\ B(v, p) &:= (\text{div } v, p)_Q, \quad B(u, q) := (\text{div } u, q)_Q. \end{aligned}$$

The operator Π_K denotes the local L^2 -projection onto the space of mean-value free functions:

$$\Pi_K \phi := \phi - \frac{1}{|K|} \int_K \phi \, dx. \quad (4.21)$$

This projection effectively removes the element-wise constant part of the test function, ensuring that the stabilization does not interfere with the pressure-velocity coupling. Furthermore, the local nature of the projection allows for a straightforward parallelization of the matrix assembly, significantly enhancing the computational efficiency for large-scale 4D problems. No additional pressure stabilization (e.g., PSPG) is included. While the P_2 – P_1 Taylor–Hood pair satisfies the discrete inf–sup condition in standard elliptic settings, its validity in this specific space–time context is treated here as a well-founded conjecture.

Discrete Problem and Mesh-Dependent Norms

The discrete problem reads: Find $(u_h, p_h) \in V_h \times Q_h$ such that

$$K_h((u_h, p_h), (v_h, q_h)) = \mathcal{F}_h(v_h, q_h) \quad \forall (v_h, q_h) \in V_h \times Q_h, \quad (4.22)$$

where the stabilized right-hand side is defined as:

$$\mathcal{F}_h(v_h, q_h) := (f, v_h)_Q + \sum_{K \in \mathcal{T}_h} \gamma_K (f, \Pi_K((w, 1) \cdot \nabla v_h))_K. \quad (4.23)$$

To analyze the stability, we equip the product space $W_h := V_h \times Q_h$ with the following mesh-dependent norms:

$$\|v_h\|_{V,h}^2 := \varepsilon \|\nabla_x v_h\|_{L^2(Q)}^2 + \sum_{K \in \mathcal{T}_h} \gamma_K \|\Pi_K((w, 1) \cdot \nabla v_h)\|_{L^2(K)}^2, \quad (4.24)$$

$$\|q_h\|_{Q,h}^2 := \|q_h\|_{L^2(Q)}^2, \quad (4.25)$$

$$\|(v_h, q_h)\|_{W,h}^2 := \|v_h\|_{V,h}^2 + \|q_h\|_{Q,h}^2. \quad (4.26)$$

4.3.2 Well-posedness

To establish the existence and uniqueness of the discrete solution, we first state the stability properties of the velocity-related bilinear form $\tilde{A}(u, v) := A(u, v) + A_{\text{supg}}(u, v)$.

Lemma 15 (Continuity and Coercivity of $\tilde{A}$) *The bilinear form $\tilde{A}(\cdot, \cdot)$ is continuous and coercive on V_h with respect to the mesh-dependent norm $\|\cdot\|_{V,h}$. In particular, there exists a constant $\alpha > 0$ such that*

$$\tilde{A}(v, v) \geq \alpha \|v\|_{V,h}^2 \quad \forall v \in \ker(B), \quad (4.27)$$

where $\ker(B) := \{v \in V_h : B(v, q) = 0, \forall q \in Q_h\}$.

Proof. The proof follows from the stability analysis of the standard SUPG formulation. For the non-projected case, it is established that

$$A(u_h, u_h) \geq \frac{1}{2} \varepsilon \|\nabla_x u_h\|_{L^2(Q)}^2 + \sum_{K \in \mathcal{T}_h} \gamma_K \|(w, 1) \cdot \nabla u_h\|_{L^2(K)}^2.$$

Since the projection Π_K is L^2 -stable ($\|\Pi_K \phi\|_{L^2(K)} \leq \|\phi\|_{L^2(K)}$), this result carries over to our formulation. The diffusive term $\varepsilon \|\nabla_x u_h\|^2$ provides control over the constant parts removed by Π_K , ensuring that $\tilde{A}(u_h, u_h) \geq \alpha \|u_h\|_{V,h}^2$. Continuity follows by applying the Cauchy–Schwarz inequality and the stability of Π_K . $\square$

Lemma 16 (Well-posedness of the Stabilized Problem) *Assume that the discrete inf-sup condition holds for the pair $V_h \times Q_h$ in the space-time setting. Since the bilinear forms $\tilde{A}(\cdot, \cdot)$ and $B(\cdot, \cdot)$ are continuous, and $\tilde{A}(\cdot, \cdot)$ is coercive on $\ker(B)$, the discrete Brezzi theorem guarantees that the stabilized problem admits a unique solution $(u_h, p_h) \in V_h \times Q_h$. Furthermore, there exists a constant $C > 0$ such that*

$$\|(u_h, p_h)\|_{W,h} \leq C \|\mathcal{F}_h\|_{W_h^*}.$$

Proof. The well-posedness is a direct application of the discrete Brezzi theorem (or the Babuška–Brezzi theory for saddle-point problems). The theorem requires:

1. Continuity of $\tilde{A}(\cdot, \cdot)$ and $B(\cdot, \cdot)$, which is satisfied due to the L^2 -stability of the projection and standard trace/inverse estimates.
2. Coercivity of $\tilde{A}(\cdot, \cdot)$ on the kernel of B , as proven in the previous Lemma.
3. The discrete inf-sup condition for $B(\cdot, \cdot)$, which is assumed to hold for the P_2 – P_1 pair.

Under these conditions, the system is stable and the unique solution (u_h, p_h) depends continuously on the data $\mathcal{F}_h$. $\square$

Lemma 17 (Consistency) *Let (u, p) be the exact solution of the continuous Oseen problem. Assuming sufficient regularity, the stabilized formulation is Galerkin-consistent:*

$$K_h((u, p), (v_h, q_h)) = \mathcal{F}_h(v_h, q_h) \quad \forall (v_h, q_h) \in V_h \times Q_h.$$

Proof. The exact solution (u, p) satisfies the strong form of the Oseen equations, which implies that the strong residual $R(u, p)$ vanishes:

$$R(u, p) := (w, 1) \cdot \nabla u - \varepsilon \Delta_x u + \nabla_x p - f = 0 \quad \text{in } Q.$$

Consequently, any weighted integral of the residual over the elements $K \in \mathcal{T}_h$ must also vanish. In our case:

$$\sum_{K \in \mathcal{T}_h} \gamma_K (R(u, p), \Pi_K((w, 1) \cdot \nabla v_h))_K = 0. \quad (4.28)$$

A key property of the projection Π_K is that it is orthogonal to constant functions in the $L^2(K)$ -sense. For any constant $c \in \mathbb{R}$, we have:

$$(\Pi_K \phi, c)_K = \int_K \left(\phi - \frac{1}{|K|} \int_K \phi \right) \cdot c \, dx = c \left(\int_K \phi \, dx - \int_K \phi \, dx \right) = 0. \quad (4.29)$$

Now consider the terms within the residual for the P_2 - P_1 discretization. On each element K :

- The Laplacian of the velocity $\Delta_x u$ for a quadratic element is constant, i.e., $\Delta_x u|_K \in [P_0]^d$.
- The gradient of the pressure $\nabla_x p$ for a linear element is constant, i.e., $\nabla_x p|_K \in [P_0]^d$.

Due to the aforementioned property of Π_K , these constant terms vanish when tested against the projection:

$$(\varepsilon \Delta_x u, \Pi_K(\phi))_K = 0 \quad \text{and} \quad (\nabla_x p, \Pi_K(\phi))_K = 0.$$

Substituting these results back into the weighted residual equation (4.28), the stabilization term simplifies to:

$$\sum_{K \in \mathcal{T}_h} \gamma_K ((w, 1) \cdot \nabla u - f, \Pi_K((w, 1) \cdot \nabla v_h))_K = 0. \quad (4.30)$$

Adding this vanishing expression to the standard weak form $A(u, v_h) + B(v_h, p) + B(u, q_h) = (f, v_h)_Q$ and rearranging the terms involving the source function f to the right-hand side yields:

$$\begin{aligned} K_h((u, p), (v_h, q_h)) &= A(u, v_h) + B(v_h, p) + B(u, q_h) \\ &\quad + \sum_{K \in \mathcal{T}_h} \gamma_K ((w, 1) \cdot \nabla u, \Pi_K((w, 1) \cdot \nabla v_h))_K \\ &= (f, v_h)_Q + \sum_{K \in \mathcal{T}_h} \gamma_K (f, \Pi_K((w, 1) \cdot \nabla v_h))_K \\ &= \mathcal{F}_h(v_h, q_h). \end{aligned}$$

This confirms the Galerkin consistency of the stabilized scheme. $\square$

4.3.3 Céa-Type-result

Theorem 7 (Céa-type Quasi-Optimality) *Assume that the stabilized bilinear form K_h is continuous with respect to the norm $\|\cdot\|_{W,h}$ with a constant*

$M > 0$, and that the discrete inf-sup condition (and thus stability of the global system) holds. Then the error satisfies:

$$\|(u - u_h, p - p_h)\|_{W,h} \leq (1 + \frac{M}{\alpha}) \inf_{(v_h, q_h) \in W_h} \|(u - v_h, p - q_h)\|_{W,h}, \quad (4.31)$$

where α is the stability constant of the discrete problem.

Remark 15 (Dependence on the diffusion parameter) *The above stability argument relies on the fact that the diffusive term $\varepsilon \|\nabla_x v_h\|_{L^2(Q)}^2$ controls the elementwise constant part removed by the projection operator Π_K . In the convection-dominated regime, however, this control becomes weak since it is scaled by the small diffusion parameter ε . Consequently, the coercivity and continuity constants of the bilinear form may depend on the ratio $\|(w, 1)\|_\infty / \varepsilon$.*

In other words, unlike the classical SUPG formulation, the present projected stabilization does not necessarily yield estimates that are uniform with respect to ε . Essentially, replacing the full streamline derivative with its projection in the stabilization term

$$\sum_K \gamma_K \int_K ((w, 1) \cdot \nabla u) \Pi_h((w, 1) \cdot \nabla v) dq$$

leads to a less effective SUPG mechanism. This projection into a smaller subspace effectively 'dampens' the stabilization, failing to provide sufficient control over the convective instabilities.

Proof. Let $(v_h, q_h) \in W_h$ be an arbitrary discrete pair. We decompose the error into an approximation error and a discrete error using the triangle inequality:

$$\|(u - u_h, p - p_h)\|_{W,h} \leq \|(u - v_h, p - q_h)\|_{W,h} + \|(v_h - u_h, q_h - p_h)\|_{W,h}. \quad (4.32)$$

From the stability of the discrete problem (guaranteed by the discrete Brezzi theorem and the inf-sup condition), there exists a constant $\alpha > 0$ such that:

$$\|(v_h - u_h, q_h - p_h)\|_{W,h} \leq \frac{1}{\alpha} \sup_{(w_h, r_h) \in W_h} \frac{K_h((v_h - u_h, q_h - p_h), (w_h, r_h))}{\|(w_h, r_h)\|_{W,h}}. \quad (4.33)$$

Applying the Galerkin orthogonality, which follows directly from the consistency lemma ($K_h(u - u_h, \cdot) = 0$), we can replace the discrete difference within the bilinear form:

$$K_h(v_h - u_h, \cdot) = K_h(v_h - u + u - u_h, \cdot) = K_h(v_h - u, \cdot).$$

Substituting this into (4.33) and utilizing the continuity of K_h (with constant M), we obtain:

$$\begin{aligned} \|(v_h - u_h, q_h - p_h)\|_{W,h} &\leq \frac{1}{\alpha} \sup_{(w_h, r_h) \in W_h} \frac{K_h(v_h - u, (w_h, r_h))}{\|(w_h, r_h)\|_{W,h}} \\ &\leq \frac{1}{\alpha} \sup_{(w_h, r_h) \in W_h} \frac{M\|v_h - u\|_{W,h}\|(w_h, r_h)\|_{W,h}}{\|(w_h, r_h)\|_{W,h}} \\ &= \frac{M}{\alpha} \|(u - v_h, p - q_h)\|_{W,h}. \end{aligned}$$

Inserting this result back into the triangle inequality (4.32) yields:

$$\|(u - u_h, p - p_h)\|_{W,h} \leq \left(1 + \frac{M}{\alpha}\right) \|(u - v_h, p - q_h)\|_{W,h}.$$

Since this inequality holds for any $(v_h, q_h) \in W_h$, taking the infimum over the discrete space completes the proof. $\square$

Final A Priori Error Estimate

To derive the global error estimate, we combine the velocity interpolation with the pressure approximation. We recall that for the P_2 – P_1 Taylor–Hood pair, the velocity is approximated with degree $k = 2$ and the pressure with degree $k - 1 = 1$.

Lemma 18 (Interpolation in the Stabilized Norm) *Let $(I_h u, J_h p)$ be the interpolants of the exact solution into $V_h \times Q_h$. Under the previously stated assumptions for the convection-dominated regime, the interpolation error in the combined norm $\|\cdot\|_{W,h}$ satisfies:*

$$\begin{aligned} \|(u - I_h u, p - J_h p)\|_{W,h}^2 &= \|u - I_h u\|_{V,h}^2 + \|p - J_h p\|_{L^2(Q)}^2 \\ &\lesssim h^{2k+1} \|(w, 1)\|_\infty \|u\|_{H^{k+1}(Q)}^2 + h^{2k} \|p\|_{H^k(Q)}^2. \end{aligned}$$

For our specific choice of $k = 2$ (quadratic velocity) and $k - 1 = 1$ (linear pressure), this yields:

$$\|(u - I_h u, p - J_h p)\|_{W,h} \lesssim h^{2.5} \|u\|_{H^3(Q)} + h^2 \|p\|_{H^2(Q)}. \quad (4.34)$$

Theorem 8 (Global A Priori Error Estimate) *Let (u, p) be the exact solution and (u_h, p_h) the stabilized discrete solution. The global error satisfies:*

$$\|(u - u_h, p - p_h)\|_{W,h} \leq C \left(h^{k+\frac{1}{2}} \|u\|_{H^{k+1}(Q)} + h^k \|p\|_{H^k(Q)} \right). \quad (4.35)$$

Specifically, for the P_2 - P_1 discretization, the convergence is limited by the pressure interpolation:

$$|||(u - u_h, p - p_h)|||_{W,h} \leq Ch^2 (h^{0.5} \|u\|_{H^3(Q)} + \|p\|_{H^2(Q)}) = \mathcal{O}(h^{2.5}). \quad (4.36)$$

Proof. The estimate follows from the Céa-type quasi-optimality (Theorem 7). The error is bounded by the best approximation in the discrete space. While the velocity component in the SUPG-norm would allow for a convergence rate of $h^{2.5}$, the L^2 -norm of the pressure interpolation error for P_1 elements is $\mathcal{O}(h^2)$. In the combined product norm, the pressure error becomes the dominant term, resulting in an overall convergence rate of $\mathcal{O}(h^2)$. $\square$

Chapter 5

Implementation of 4D FEM

As discussed in the previous chapters, we consider *space-time continuous numerical methods*. This approach leads to a fundamental and concrete challenge: if the spatial domain is three-dimensional, then the resulting space-time problem becomes four-dimensional. Consequently, the numerical discretization requires *four-dimensional finite elements* as well as a corresponding *four-dimensional mesh*.

To the best of the author's knowledge, the solution of such genuinely four-dimensional finite element problems is currently not supported by publicly available finite element libraries, such as `NGSolve`. This naturally raises the question of how such four-dimensional finite elements and meshes can be constructed and implemented in practice. In advance, it should be noted that the author has implemented a dedicated four-dimensional finite element framework from scratch. A reference to the corresponding GitHub repository will be provided. The purpose of this chapter is to present the key mathematical concepts and results that underpin this implementation. In particular, we focus on the construction of four-dimensional simplicial elements, the associated mappings to reference elements, and the geometric quantities required for numerical integration and finite element assembly.

5.1 4D Semi-Structured Mesh

There exist several approaches for constructing a four-dimensional mesh. In general, these approaches can be classified into *structured* and *unstructured* mesh constructions, each of which comes with distinct advantages and disadvantages.

Structured meshes are typically easy to generate and allow for a very regular data layout, which facilitates efficient memory access and straight-

forward implementation of numerical algorithms. Moreover, their regular structure often simplifies the definition of connectivity and element neighborhoods. However, structured meshes are severely limited when it comes to representing complex geometries or local mesh refinement, which are essential in many practical applications. Unstructured meshes, on the other hand, offer a high degree of geometric flexibility and allow for local refinement in regions where increased resolution is required. This makes them particularly well-suited for complex spatial domains. The main drawbacks of unstructured meshes are their more complicated data structures, higher implementation complexity, and increased computational overhead in mesh traversal and assembly. In this work, we adopt a *hybrid* or *semi-structured* approach. Starting from an unstructured mesh in three-dimensional space, we extend the mesh in the temporal direction to obtain a four-dimensional space–time mesh. More precisely, each three-dimensional spatial element is extruded in time, resulting in a four-dimensional prism. This prism may consist of several temporal layers, corresponding to different time slabs.

The crucial step in the construction is the subdivision of each four-dimensional prism into four-dimensional simplicial elements, so-called *pentatopes*. Consider a tetrahedron at time level t_n with vertices x_1, x_2, x_3, x_4 , and its corresponding tetrahedron at time level t_{n+1} with vertices y_1, y_2, y_3, y_4 . The resulting space–time prism can be decomposed into four pairwise disjoint pentatopes given by

$$(x_1, x_2, x_3, x_4, y_4), \quad (x_1, x_2, x_3, y_4, y_3), \quad (x_1, x_2, y_4, y_3, y_2), \quad (x_1, y_4, y_3, y_2, y_1).$$

This decomposition, proposed for example in [17], yields a conforming four-dimensional simplicial mesh that is well-suited for the application of space–time finite element methods.

5.1.1 Decomposition of a space–time prism into four pentatopes

Let $K_n = \text{conv}\{x_1, x_2, x_3, x_4\}$ be a tetrahedron at time t_n and let $K_{n+1} = \text{conv}\{y_1, y_2, y_3, y_4\}$ be its counterpart at time t_{n+1} . The corresponding space–time prism is the convex hull

$$Q := \text{conv}\{x_1, x_2, x_3, x_4, y_1, y_2, y_3, y_4\} \subset \mathbb{R}^4.$$

We consider the four pentatopes

$$\begin{aligned} P_1 &= \text{conv}(x_1, x_2, x_3, x_4, y_4), & P_2 &= \text{conv}(x_1, x_2, x_3, y_4, y_3), \\ P_3 &= \text{conv}(x_1, x_2, y_4, y_3, y_2), & P_4 &= \text{conv}(x_1, y_4, y_3, y_2, y_1). \end{aligned}$$

Reference configuration. We prove disjointness and covering on a reference prism and then transfer the result to Q by an affine space–time mapping.

Let $\Delta^3 := \{\lambda \in \mathbb{R}^4 : \lambda_i \geq 0, \sum_{i=1}^4 \lambda_i = 1\}$ be the reference tetrahedron (in barycentric coordinates) and define the reference prism

$$\widehat{Q} := \Delta^3 \times [0, 1].$$

Denote the tetrahedron vertices by $v_1, \dots, v_4$. We identify bottom and top vertices of the prism as

$$\widehat{x}_i := (v_i, 0), \quad \widehat{y}_i := (v_i, 1), \quad i = 1, \dots, 4.$$

Define four reference pentatopes by

$$\begin{aligned} \widehat{P}_1 &= \text{conv}(\widehat{x}_1, \widehat{x}_2, \widehat{x}_3, \widehat{x}_4, \widehat{y}_4), \\ \widehat{P}_2 &= \text{conv}(\widehat{x}_1, \widehat{x}_2, \widehat{x}_3, \widehat{y}_4, \widehat{y}_3), \\ \widehat{P}_3 &= \text{conv}(\widehat{x}_1, \widehat{x}_2, \widehat{y}_4, \widehat{y}_3, \widehat{y}_2), \\ \widehat{P}_4 &= \text{conv}(\widehat{x}_1, \widehat{y}_4, \widehat{y}_3, \widehat{y}_2, \widehat{y}_1). \end{aligned}$$

Lemma 19 (Partition of the reference prism) *The interiors of $\widehat{P}_1, \widehat{P}_2, \widehat{P}_3, \widehat{P}_4$ are pairwise disjoint and*

$$\widehat{Q} = \widehat{P}_1 \cup \widehat{P}_2 \cup \widehat{P}_3 \cup \widehat{P}_4.$$

Moreover, each point $(\lambda, \tau) \in \widehat{Q}$ belongs to exactly one $\widehat{P}_k$ unless it lies on an interface.

Proof. Take an arbitrary point $(\lambda, \tau) \in \widehat{Q}$, i.e. $\lambda \in \Delta^3$ and $\tau \in [0, 1]$. We show that membership in the simplices is characterized by simple inequalities in τ .

Step 1: Characterization of $\widehat{P}_1$. A point lies in $\widehat{P}_1$ iff it can be written as a convex combination of $\widehat{x}_1, \widehat{x}_2, \widehat{x}_3, \widehat{x}_4, \widehat{y}_4$:

$$(\lambda, \tau) = \alpha_1 \widehat{x}_1 + \alpha_2 \widehat{x}_2 + \alpha_3 \widehat{x}_3 + \alpha_4 \widehat{x}_4 + \alpha_5 \widehat{y}_4, \quad \alpha_i \geq 0, \quad \sum_{i=1}^5 \alpha_i = 1.$$

Since only $\widehat{y}_4$ has time coordinate 1, we have $\tau = \alpha_5$. In barycentric coordinates this gives $\lambda_1 = \alpha_1$, $\lambda_2 = \alpha_2$, $\lambda_3 = \alpha_3$ and

$$\lambda_4 = \alpha_4 + \alpha_5 \Rightarrow \tau = \alpha_5 \leq \lambda_4.$$

Conversely, if $\tau \leq \lambda_4$, set $\alpha_5 = \tau$, $\alpha_4 = \lambda_4 - \tau$, and $\alpha_i = \lambda_i$ for $i = 1, 2, 3$. All coefficients are nonnegative and sum to 1, hence $(\lambda, \tau) \in \widehat{P}_1$. Therefore,

$$(\lambda, \tau) \in \widehat{P}_1 \iff 0 \leq \tau \leq \lambda_4.$$

Step 2: Characterization of $\widehat{P}_2, \widehat{P}_3, \widehat{P}_4$. Proceeding analogously, write (λ, τ) as a convex combination of the vertices of each simplex:

- For $\widehat{P}_2 = \text{conv}(\widehat{x}_1, \widehat{x}_2, \widehat{x}_3, \widehat{y}_4, \widehat{y}_3)$, one obtains the necessary and sufficient conditions

$$(\lambda, \tau) \in \widehat{P}_2 \iff \lambda_4 \leq \tau \leq \lambda_3 + \lambda_4.$$

- For $\widehat{P}_3 = \text{conv}(\widehat{x}_1, \widehat{x}_2, \widehat{y}_4, \widehat{y}_3, \widehat{y}_2)$, one obtains

$$(\lambda, \tau) \in \widehat{P}_3 \iff \lambda_3 + \lambda_4 \leq \tau \leq \lambda_2 + \lambda_3 + \lambda_4.$$

- For $\widehat{P}_4 = \text{conv}(\widehat{x}_1, \widehat{y}_4, \widehat{y}_3, \widehat{y}_2, \widehat{y}_1)$, one obtains

$$(\lambda, \tau) \in \widehat{P}_4 \iff \lambda_2 + \lambda_3 + \lambda_4 \leq \tau \leq 1.$$

Step 3: Disjointness and covering. Since $\lambda_i \geq 0$ and $\sum_{i=1}^4 \lambda_i = 1$, the quantities

$$0 \leq \lambda_4 \leq \lambda_3 + \lambda_4 \leq \lambda_2 + \lambda_3 + \lambda_4 \leq 1$$

form an ordered partition of the time interval $[0, 1]$. Hence every $(\lambda, \tau) \in \widehat{Q}$ satisfies exactly one of the four sets of inequalities, so it belongs to (at least) one $\widehat{P}_k$, proving $\widehat{Q} = \bigcup_{k=1}^4 \widehat{P}_k$. Moreover, if τ lies strictly between the bounds, the membership is unique, which implies that the interiors are pairwise disjoint. Intersections can only occur when τ equals one of the interface values λ_4 , $\lambda_3 + \lambda_4$, or $\lambda_2 + \lambda_3 + \lambda_4$, i.e. on lower-dimensional faces. $\square$

Transfer to the physical prism. Define the (bi-affine in (λ, τ) , and affine on each $\widehat{P}_k$) space-time mapping

$$F(\lambda, \tau) := \sum_{i=1}^4 \lambda_i ((1 - \tau)x_i + \tau y_i) \in \mathbb{R}^4.$$

Then $F(\widehat{x}_i) = x_i$ and $F(\widehat{y}_i) = y_i$, hence $F(\widehat{Q}) = Q$. Furthermore, for each k the restriction $F|_{\widehat{P}_k}$ is affine, so

$$F(\widehat{P}_1) = P_1, \quad F(\widehat{P}_2) = P_2, \quad F(\widehat{P}_3) = P_3, \quad F(\widehat{P}_4) = P_4.$$

Since F is one-to-one on the prism and preserves convexity, the disjointness of interiors and the covering property transfer from the reference setting to the physical one. Therefore,

$$Q = P_1 \cup P_2 \cup P_3 \cup P_4, \quad \text{int}(P_i) \cap \text{int}(P_j) = \emptyset \text{ for } i \neq j,$$

and any overlap can occur only on shared faces.

A key computational advantage is that the Jacobian $J_k = \nabla F|_{\hat{P}_k}$ is constant on each sub-element. For a simplex P_k with vertices $\{v_0, v_1, v_2, v_3, v_4\}$, it is efficiently computed as the concatenation of the edge vectors:

$$J_k = [v_1 - v_0, \quad v_2 - v_0, \quad v_3 - v_0, \quad v_4 - v_0] \in \mathbb{R}^{4 \times 4}.$$

5.2 Numerical Integration on 4D Simplices

A fundamental component in the implementation of the Finite Element Method (FEM) is the accurate evaluation of integrals over the reference element. Since our formulation utilizes 4D simplices (pentatopes), choosing an appropriate quadrature rule is essential for both precision and computational efficiency. In the following, we discuss two distinct strategies for numerical integration on 4D simplices.

5.2.1 Grundmann–Möller Integration Rule

The Grundmann–Möller rule provides an explicit construction for invariant integration formulas on n -dimensional simplices for any given degree of polynomial exactness d .

Theorem 9 *Let $s \in \mathbb{N}_0$, $d = 2s + 1$, and $n \in \mathbb{N}$ be the dimension of the simplex. The formula*

$$Q[f] = \sum_{i=0}^s (-1)^i \frac{2^{-2s}}{i!(n+s-i)!} \frac{(n+d-2i)^{d+n}}{(d+n-i)!} \sum_{|\beta|=s-i} f\left(\frac{2\beta+1}{n+d-2i}\right) \quad (5.1)$$

is an invariant integration formula of degree d .

A rigorous proof and the derivation of the weights can be found in the work of Grundmann and Möller [12]. This rule is particularly advantageous as it provides an explicit formula for a simplex of any order. In the context of this thesis, this rule was utilized to ensure high-order convergence for the space-time integrals and was accessed via existing Python libraries for the numerical implementation.

5.2.2 Duffy Transformation

An alternative strategy for numerical integration is the so-called *Duffy Transformation*. Instead of defining a quadrature rule directly on the simplex, this method utilizes a mapping from a simpler reference domain.

The core principle of the Duffy Transformation is to map a hypercube (or a product of intervals $[0, 1]^n$) onto a simplex. In four dimensions, this is achieved by a sequence of non-linear transformations that "collapse" certain faces of the hypercube into the vertices of the simplex. The mapping $\mathcal{T} : [0, 1]^4 \rightarrow \hat{P}$ is defined as:

$$\begin{aligned} x_1 &= \xi_1 \\ x_2 &= (1 - \xi_1)\xi_2 \\ x_3 &= (1 - \xi_1)(1 - \xi_2)\xi_3 \\ x_4 &= (1 - \xi_1)(1 - \xi_2)(1 - \xi_3)\xi_4 \end{aligned}$$

where $(\xi_1, \dots, \xi_4)$ are the coordinates in the unit hypercube. The Duffy transformation $\mathcal{T}$ allows us to evaluate an integral over the 4D reference simplex $\hat{P}$ by transforming it into an integral over the 4D unit hypercube $\hat{C} = [0, 1]^4$. The transformation is given by the change of variables:

$$\int_{\hat{P}} f(x_1, x_2, x_3, x_4) dx = \int_0^1 \int_0^1 \int_0^1 \int_0^1 f(\mathcal{T}(\xi)) |J_{\mathcal{T}}(\xi)| d\xi_4 d\xi_3 d\xi_2 d\xi_1 \quad (5.2)$$

where $|J_{\mathcal{T}}(\xi)|$ is the determinant of the Jacobian matrix of the mapping. For the 4D Duffy transformation, the determinant is:

$$|J_{\mathcal{T}}(\xi)| = (1 - \xi_1)^3 (1 - \xi_2)^2 (1 - \xi_3)^1. \quad (5.3)$$

Substituting the mapping and the Jacobian, the explicit integration formula becomes:

$$\int_{\hat{P}} f dx = \int_0^1 (1 - \xi_1)^3 \int_0^1 (1 - \xi_2)^2 \int_0^1 (1 - \xi_3) \int_0^1 f(\mathcal{T}(\xi_1, \xi_2, \xi_3, \xi_4)) d\xi_4 d\xi_3 d\xi_2 d\xi_1. \quad (5.4)$$

Numerical Implementation: In practice, this allows for the application of a one-dimensional Gauss–Legendre quadrature rule in each direction. For a rule with N points and weights $\{w_j, \zeta_j\}_{j=1}^N$, the integral is approximated as:

$$\int_{\hat{P}} f dx \approx \sum_{i=1}^N \sum_{j=1}^N \sum_{k=1}^N \sum_{l=1}^N w_i w_j w_k w_l \cdot \mathcal{J}(\zeta_i, \zeta_j, \zeta_k) \cdot f(\mathcal{T}(\zeta_i, \zeta_j, \zeta_k, \zeta_l)), \quad (5.5)$$

where $\mathcal{J}$ represents the precomputed Jacobian weights. Note that while this method is conceptually simple and easy to implement using tensor products, it requires N^4 function evaluations, which makes it computationally more expensive than the Grundmann–Möller rule for higher dimensions.

Advantages and Characteristics:

- **Tensor Product Quadrature:** It allows the use of standard, highly optimized Gauss–Legendre quadrature rules defined for cubes.
- **Jacobian Determinant:** The transformation introduces a singularity in the Jacobian at the collapsed vertices. However, this is analytically compensated by the determinant of the transformation, which effectively clusters the integration points near the vertices.
- **Flexibility:** While the Grundmann–Möller rule is often more efficient in terms of the number of points for a given degree, the Duffy transformation is very flexible when dealing with singular integrands.

Chapter 6

Numerical Results

In this chapter we are going to present some numerical results, showing the workability of the different methods presented in the analytical chapter. We will concentrate on two important properties. We will on the one hand compare the stabilized SUPG-method to the standard continuous Galerkin space-time FEM. This should highlight once again the necessity of the SUPG stabilization. In addition we will test the convergence order and see if it matches the analytical result.

6.1 Scalar Advection Diffusion

6.1.1 Stability Comparison in 2D

As a guidance example, we consider the time-dependent convection–diffusion problem

$$\partial_t u(x, t) + w \cdot \nabla_x u(x, t) - \Delta_x u(x, t) = f(x, t) \quad \text{in } Q := \Omega \times (0, T), \quad (6.1)$$

with Dirichlet boundary data and initial conditions prescribed by the analytically constructed solution

$$u_D(x, t) = u_{\text{exact}}(x, t), \quad (x, t) \in \partial\Omega \times (0, T), \quad (6.2)$$

$$u(x, 0) = u_{\text{exact}}(x, 0), \quad x \in \Omega, \quad (6.3)$$

We choose $\Omega = (0, 1)$ and $T = 1$, hence $Q = (0, 1) \times (0, 1)$. For a given diffusion parameter $\varepsilon > 0$, the exact solution is defined as

$$u_{\text{exact}}(x, t) = \frac{1 - \exp\left(\frac{x-1}{\varepsilon}\right)}{1 - \exp\left(-\frac{1}{\varepsilon}\right)} \exp(-t), \quad (6.4)$$

and the right-hand side is chosen accordingly as

$$f(x, t) = -\frac{1 - \exp\left(\frac{x-1}{\varepsilon}\right)}{1 - \exp\left(-\frac{1}{\varepsilon}\right)} \exp(-t). \quad (6.5)$$

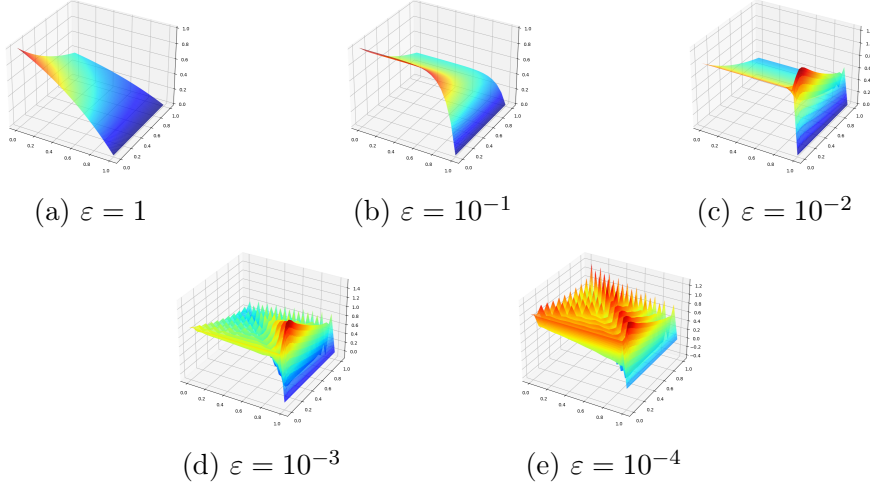


Figure 6.1: Unstabilized numerical solutions of the advection diffusion problem for decreasing diffusion parameters ε . Strong non-physical oscillations occur in the convection-dominated regime.

In this section, we once again emphasize the necessity of SUPG stabilization in convection-dominated regimes. To this end, we consider the limiting case $\varepsilon \rightarrow 0$ and investigate the numerical solution of the advection diffusion equation *without* any stabilization. Figure 6.1 shows the numerical solutions obtained for decreasing values of the diffusion parameter ε . As the diffusion becomes small compared to the advective transport, the standard Galerkin finite element method fails to produce physically meaningful solutions. Instead, strong non-physical oscillations appear throughout the computational domain, in particular in regions where sharp layers are present. These oscillations are a well-known problem of convection-dominated PDEs and were already discussed theoretically in Chapter 2. In contrast, Figure 6.2 clearly demonstrates the beneficial effect of SUPG stabilization. The stabilized solution no longer exhibits non-physical oscillations, and no spurious over- or undershoots can be observed, even in the convection-dominated regime. The solution remains smooth and physically meaningful throughout the domain.

This qualitative observation is further supported by the quantitative results reported in Table 6.1. While the L^2 -error of the unstabilized formulation

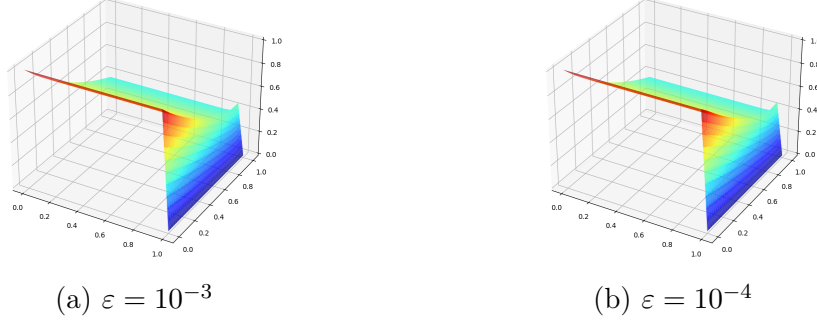


Figure 6.2: SUPG-stabilized numerical solutions of the advection diffusion problem for small diffusion parameters ε . No spurious oscillations are observed in contrast to the unstabilized formulation.

increases significantly as $\varepsilon \rightarrow 0$, indicating a loss of robustness, the SUPG-stabilized method maintains comparable error levels over the entire range of diffusion parameters. In particular, for very small values of ε , the stabilized solution does not exhibit any error growth, confirming the effectiveness of the SUPG approach in convection-dominated regimes.

Table 6.1: L^2 -errors for the advection diffusion problem with and without SUPG stabilization for different values of the diffusion parameter ε .

ε	L^2 -error (no SUPG)	L^2 -error (SUPG)
10^{-4}	2.55061×10^{-1}	8.4496×10^{-2}
10^{-3}	1.79310×10^{-1}	8.4602×10^{-2}
10^{-2}	5.0538×10^{-2}	5.8975×10^{-2}
10^{-1}	2.916×10^{-3}	1.8505×10^{-2}
1	2.49×10^{-4}	9.02×10^{-4}

6.1.2 Stability Comparison in Four Space-Time Dimensions

As a guiding example, we consider the four-dimensional space-time advection diffusion problem with an analytically constructed solution. The exact solution is given by

$$u_{\text{ex}}(x, t) = \frac{1 - \exp\left(\frac{x_1 + x_2 + x_3 - 3}{\varepsilon}\right)}{1 - \exp\left(-\frac{1}{\varepsilon}\right)} \exp(-t),$$

where $x = (x_1, x_2, x_3) \in \Omega = [0, 1]^3$ and $t \in (0, 1]$. The right-hand side f is chosen consistently as

$$f(x, t) = -\frac{1 - \exp\left(\frac{x_1 + x_2 + x_3 - 3}{\varepsilon}\right)}{1 - \exp\left(-\frac{1}{\varepsilon}\right)} \exp(-t).$$

As already observed in the two-dimensional setting, the advection diffusion problem becomes increasingly challenging for small diffusion parameters ε . This behavior carries over directly to the four-dimensional space-time formulation. In particular, for convection-dominated regimes, unstabilized finite element discretizations exhibit strong oscillations and a severe loss of accuracy.

Table 6.2 reports the L^2 -difference between the numerical and exact solution for decreasing values of ε without stabilization. A dramatic increase of the error can be observed as ε decreases, indicating a complete breakdown of the standard Galerkin method in the convection-dominated regime. These results clearly indicate that SUPG stabilization is not only beneficial but essential in four-dimensional space-time simulations. Despite the increased dimensionality, the stabilization mechanism remains effective and successfully suppresses spurious oscillations. Consequently, the numerical experiments confirm that the space-time SUPG formulation extends naturally from lower-dimensional problems to the fully four-dimensional setting.

Table 6.2: L^2 -error comparison for the four-dimensional advection diffusion problem with and without SUPG stabilization.

ε	L^2 -error (no SUPG)	L^2 -error (SUPG)
10^{-7}	9.66415×10^1	4.98098×10^{-1}
10^{-6}	9.64398×10^0	5.1212×10^{-2}
10^{-5}	9.45554×10^{-1}	1.2802×10^{-2}
10^{-4}	8.96456×10^{-2}	1.1786×10^{-2}

6.2 Convergence Study

In this section, we investigate the convergence behavior of the proposed space-time SUPG discretization. The convergence order is measured in the L^2 -norm by comparing numerical solutions obtained on successively refined meshes. The experimental order of convergence is computed between two mesh sizes h and $h/2$. We first consider finite element spaces consisting of polynomials of degree $k = 2$. According to the theoretical error estimate

derived in the previous sections, we expect an L^2 -convergence rate of order $k + \frac{1}{2} = 2.5$. As shown in Table 6.3, the experimentally observed convergence

Table 6.3: Experimental L^2 -convergence order for polynomial degree $k = 2$.

Mesh refinement	Convergence order
$h = 0.5 \rightarrow 0.25$	2.7265
$h = 0.25 \rightarrow 0.125$	1.9389
$h = 0.125 \rightarrow 0.0625$	2.2349
$h = 0.0625 \rightarrow 0.03125$	1.4687
$h = 0.03125 \rightarrow 0.015625$	3.1178
$h = 0.015625 \rightarrow 0.0078125$	2.4147
Average	2.3169

orders fluctuate around the theoretically predicted value. The average convergence order of approximately 2.32 is in good agreement with the expected rate of $k + \frac{1}{2} = 2.5$. The observed variations are typical for convection-dominated problems and can be attributed to pre-asymptotic effects. To further support these findings, we repeat the convergence study for polynomial degree $k = 1$. In this case, the theoretical convergence rate in the L^2 -norm is $k + \frac{1}{2} = 1.5$. Table 6.4 confirms the theoretical prediction for linear finite

Table 6.4: Experimental L^2 -convergence order for polynomial degree $k = 1$.

Mesh refinement	Convergence order
$h = 0.5 \rightarrow 0.25$	1.5021
$h = 0.25 \rightarrow 0.125$	1.5018
$h = 0.125 \rightarrow 0.0625$	2.1985
$h = 0.0625 \rightarrow 0.03125$	1.8558
$h = 0.03125 \rightarrow 0.015625$	2.4069
$h = 0.015625 \rightarrow 0.0078125$	1.6950
Average	1.8600

elements. The average convergence order of approximately 1.86 is close to the expected rate of 1.5. Again, small deviations can be explained by non-asymptotic effects on coarse meshes. All convergence studies are performed for the smooth test problem

$$u_{\text{ex}}(x, t) = \sin(\pi(x - wt)),$$

with the right-hand side f constructed by the method of manufactured solutions. Due to the cancellation of the advection terms, the forcing function

reduces essentially to a diffusion contribution. Consequently, the exact solution is smooth and possesses the regularity required for optimal convergence rates. This observation is crucial when comparing the present results with earlier numerical experiments. For problems with very small diffusion parameters $\varepsilon \rightarrow 0$, the solution typically develops sharp boundary or interior layers. These layers cannot be resolved by the mesh and effectively introduce discontinuities. As a result, the exact solution lacks sufficient regularity and belongs only to $L^2(\Omega)$. In such cases, convergence rates are limited to order 0.5, independently of the polynomial degree.

6.3 Oseen equation

We consider the unsteady Oseen equations as a linearized model for the Navier–Stokes equations. Let $\tilde{Q} := \Omega \times (0, T)$ denote the space-time domain with the lateral boundary $\Sigma := \partial\Omega \times (0, T)$. Given a divergence-free convection field $w : \tilde{Q} \rightarrow \mathbb{R}^d$ (with $\nabla_x \cdot w = 0$), a source term f , Dirichlet data u_D , and an initial condition u_0 , the problem seeks a velocity field u and a pressure p such that:

$$\partial_t u - \nu \Delta_x u + (w \cdot \nabla_x)u + \nabla_x p = f \quad \text{in } \tilde{Q}, \quad (6.6a)$$

$$\nabla_x \cdot u = 0 \quad \text{in } \tilde{Q}, \quad (6.6b)$$

$$u = u_D \quad \text{on } \Sigma, \quad (6.6c)$$

$$u(\cdot, 0) = u_0 \quad \text{in } \Omega. \quad (6.6d)$$

6.3.1 Problem Setup and Analytical Reference Solution

To validate the numerical scheme, we employ the method of manufactured solutions for the Oseen problem. We define an analytical solution $(\mathbf{u}_{ex}, p_{ex})$ on the space-time domain $\Omega \times I$. The chosen velocity field $\mathbf{u} = (u_1, u_2, u_3)^T$ is given by:

$$u_1(x, y, z, t) = e^{-t} \sin(k\pi x) \cos(k\pi y) \cos(k\pi z) \quad (6.7)$$

$$u_2(x, y, z, t) = -\frac{1}{2} e^{-t} \cos(k\pi x) \sin(k\pi y) \cos(k\pi z) \quad (6.8)$$

$$u_3(x, y, z, t) = -\frac{1}{2} e^{-t} \cos(k\pi x) \cos(k\pi y) \sin(k\pi z) \quad (6.9)$$

where k is a wave number constant. The exact pressure is set to $p_{ex} = 0$. By substituting these expressions into the Oseen operator $\mathcal{L}(\mathbf{u}, p) = \partial_t \mathbf{u} +$

$(\mathbf{w} \cdot \nabla)\mathbf{u} - \nu \Delta \mathbf{u} + \nabla p$, we derive the force term $\mathbf{f} = (f_1, f_2, f_3)^T$. Defining the diffusion constant $C_{diff} = 3(k\pi)^2\nu$, the components of the source term are:

$$\begin{aligned}
f_1 = e^{-t} & \left[\underbrace{-\sin(k\pi x) \cos(k\pi y) \cos(k\pi z)}_{\text{time derivative } \partial_t u_1} \right. \\
& + k\pi \cdot w_x \cos(k\pi x) \cos(k\pi y) \cos(k\pi z) \\
& - k\pi \cdot w_y \sin(k\pi x) \sin(k\pi y) \cos(k\pi z) \\
& - k\pi \cdot w_z \sin(k\pi x) \cos(k\pi y) \sin(k\pi z) \\
& \left. + \underbrace{C_{diff} \sin(k\pi x) \cos(k\pi y) \cos(k\pi z)}_{\text{diffusion } -\nu \Delta u_1} \right] \tag{6.10}
\end{aligned}$$

(The components f_2 and f_3 are derived analogously). This setup ensures that the divergence-free constraint $\nabla \cdot \mathbf{u} = 0$ is satisfied. The numerical errors presented in the following sections are calculated as the L^2 -norm of the difference between these analytical functions and the discrete finite element solution.

Computational Domain and Parameters For the numerical simulations presented in this chapter, we consider the specific spatial domain $\Omega := (0, 1)^3$ and a final time $T = 1$. Consequently, the space-time cylinder is given by $\tilde{Q} = (0, 1)^3 \times (0, 1)$. The computations were performed with a mesh refinement parameter of $h = 0.25$. These geometric parameters remain constant across all test cases to ensure the comparability of the L^2 -error magnitudes.

6.3.2 P_2/P_1 Method and Projected SUPG

In this section, we solve the convection-dominated Oseen equation using the classical Taylor–Hood P_2/P_1 elements. We compare the standard Galerkin approach to the stabilized projected SUPG method for P2-P1 elements presented in Chapter 4.

As shown in Table 6.5, the projected SUPG method produces slightly improved results compared to the unstabilized case for $\nu = 10^{-5}$. However, the method clearly lacks the necessary robustness for this regime. This deficiency stems from the issue previously addressed in our remark: by projecting the streamline term $(w, 1) \cdot \nabla v$, we lose the essential "upwinding" property inherent in the original SUPG formulation.

The projection effectively weakens the streamline derivative, making the stabilization significantly less effective. This loss of control over the convective gradients becomes even more apparent when compared to the stabilized

P_1/P_1 results in the following section. There, the standard SUPG method achieves errors of order 10^{-1} , whereas the P_2/P_1 projected approach remains stuck at errors of order 10^1 , demonstrating a much poorer performance in the high-Péclet limit.

Table 6.5: Comparison of L^2 -velocity errors for $\nu = 10^{-5}$ using P_2/P_1 Taylor–Hood elements: Unstabilized vs. Stabilized with Projected SUPG.

Component	L^2 -Error	Component	L^2 -Error
$\ u_1 - u_{1,h}\ $	19.1690	$\ u_1 - u_{1,h}\ $	13.0305
$\ u_2 - u_{2,h}\ $	19.3125	$\ u_2 - u_{2,h}\ $	12.9229
$\ u_3 - u_{3,h}\ $	19.2663	$\ u_3 - u_{3,h}\ $	13.5320
(a) Unstabilized P_2/P_1		(b) Projected SUPG (P_2/P_1)	

Table 6.6: Comparison of L^2 -velocity errors for the case ($\nu = 1$) using P_2/P_1 Taylor–Hood elements.

Component	L^2 -Error	Component	L^2 -Error
$\ u_1 - u_{1,h}\ $	0.1824	$\ u_1 - u_{1,h}\ $	0.2414
$\ u_2 - u_{2,h}\ $	0.0809	$\ u_2 - u_{2,h}\ $	0.1117
$\ u_3 - u_{3,h}\ $	0.1284	$\ u_3 - u_{3,h}\ $	0.0840
(a) Unstabilized P_2/P_1 ($\nu = 1$)		(b) Projected SUPG P_2/P_1 ($\nu = 1$)	

6.3.3 Numerical Results for the P1-P1 Method

In the following, the Oseen problem is analyzed using a space-time discretization with P_1/P_1 finite elements. We compare a method stabilized only by PSPG (Pressure-Stabilizing/Petrov-Galerkin) with our fully stabilized SUPG/PSPG method.

Table 6.7 shows the L^2 -errors for the variant using only a PSPG stabilization of the form $\delta \int_K \nabla p \cdot \nabla q \, dx$ (scaled with h^2). In contrast, Table 6.8 presents the fully stabilized method, including the SUPG term for the momentum equation.

As shown in the tables, the method without full stabilization (lacking SUPG) yields significantly worse L^2 -errors for convection-dominated problems ($\nu \rightarrow 0$) compared to the fully stabilized variant. While the error for the PSPG-only method stagnates at approximately 0.6 for small viscosities,

Table 6.7: L^2 -errors of the P_1/P_1 discretization with PSPG stabilization only.

ν	$\ u_1 - u_{1,h}\ _{L^2}$	$\ u_2 - u_{2,h}\ _{L^2}$	$\ u_3 - u_{3,h}\ _{L^2}$	$\ p - p_h\ _{L^2}$
1	0.1005	0.0473	0.0508	0.0491
10^{-5}	0.6563	0.6913	0.5207	0.1373
10^{-6}	0.6479	0.6822	0.5148	0.1341
10^{-7}	0.6471	0.6814	0.5142	0.1338

Table 6.8: L^2 -errors of the P_1/P_1 discretization with SUPG and PSPG stabilization.

ν	$\ u_1 - u_{1,h}\ _{L^2}$	$\ u_2 - u_{2,h}\ _{L^2}$	$\ u_3 - u_{3,h}\ _{L^2}$	$\ p - p_h\ _{L^2}$
1	0.12026893	0.06016142	0.06420589	0.13187630
10^{-5}	0.10418531	0.04839712	0.05214461	0.05094262
10^{-6}	0.10418618	0.04839742	0.05214498	0.05094259
10^{-7}	0.10418627	0.04839745	0.05214501	0.05094259

the additional SUPG stabilization reduces the error to approximately 0.05 to 0.1. This corresponds to an improvement in accuracy by more than a factor of 10.

Nevertheless, it is noteworthy that the PSPG stabilization alone prevents the errors from exploding and positively influences the matrix conditioning. However, our proposed SUPG method captures the physical gradients much more precisely, making it the superior choice for the Oseen problem considered here. The fact that the errors in Table 6.8 only vary in the fifth or sixth decimal place confirms that the method has reached the "convection-limit" for the given mesh, where further reductions in the physical viscosity ν no longer significantly alter the numerical representation of the velocity and pressure fields. It is crucial to note that, especially when contrasted with the unstabilized Taylor–Hood elements, the proposed stabilization proves indispensable to prevent numerical oscillations from compromising the results.

Chapter 7

Conclusion

In this thesis, I investigated a stabilized space-time finite element method based on the Streamline-Upwind/Petrov-Galerkin (SUPG) approach for the scalar convection-diffusion equation. I demonstrated that the standard Galerkin method exhibits well-known instabilities, particularly in the convection-dominated regime, which necessitates the use of stabilization techniques. Such methods are not only vital for scalar problems but also serve as a foundation for more complex systems, such as the Oseen equations. I presented three distinct discretizations, all of them employing the SUPG framework to enhance approximation quality. For all methods discussed, a rigorous a priori error analysis was conducted, providing theoretical convergence rates that were subsequently validated by numerical experiments. Furthermore, I addressed the practical implementation by proposing and implementing several strategies for the construction of efficient 4D finite element solvers. A key finding of this study is the significant performance difference between the two discretizations evaluated for the Oseen problem. The P_1/P_1 stabilization proved to be far superior, as it accounts for the full SUPG effect. In contrast, the projected stabilization approach suffers from the fact that the projection operator essentially "breaks" the stabilization's effectiveness. By projecting the streamline term into a lower-dimensional subspace, the crucial upwind property is diminished, leading to the loss of robustness in the convection-dominated regime. Therefore, for future developments of space-time SUPG methods, it is imperative to focus specifically on the consistent treatment of the streamline-diffusion term:

$$\sum_K \gamma_K \int_K ((w, 1) \cdot \nabla u) ((w, 1) \cdot \nabla v) \, dq. \quad (7.1)$$

Maintaining the full strength of this term without excessive projection or modification is the key to ensuring stability and high-order convergence in

high-Péclet flows.

Outlook

A natural extension of this work would be the application of the SUPG stabilization to the full incompressible Navier-Stokes equations. Moreover, as the inf-sup stability for the space-time Taylor-Hood element was assumed as a conjecture within this framework, a formal proof of this stability criterion remains an important open mathematical challenge for future research.

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